

原文献表格合集

SEU 旋转机械故障诊断 · 对照用原文表格页

- 改进Transformer (Guan, ACM 2023) SEU齿轮5类·80/20·SNR10dB
- ICEEMDAN-MPE-AWT (Gao, Appl Sci 2024) SEU齿轮5类
- SVM熵+机器学习 (Zhang, Algorithms 2023) SEU轴承/齿轮5类
- FE-MCFormer (arXiv 2025) ※ PU+压缩机数据集, 非SEU
- LECA-EfficientNetV2 (Cheng, Sci Rep 2024, 14:743) SEU轴承/齿轮5类
- SCBM-Net (Liu, Sci Rep 2025, 15:37904) SEU轴承10类
- SAGAN-IResNet (Wang, Sci Rep 2025) SEU齿轮5类

说明：每页为对应论文 PDF 的原始页面截图（含表号与原文数值）。

注意：FE-MCFormer 经核对为 PU (Paderborn) + 压缩机数据集, 非 SEU, 仅作强噪声对比参考。

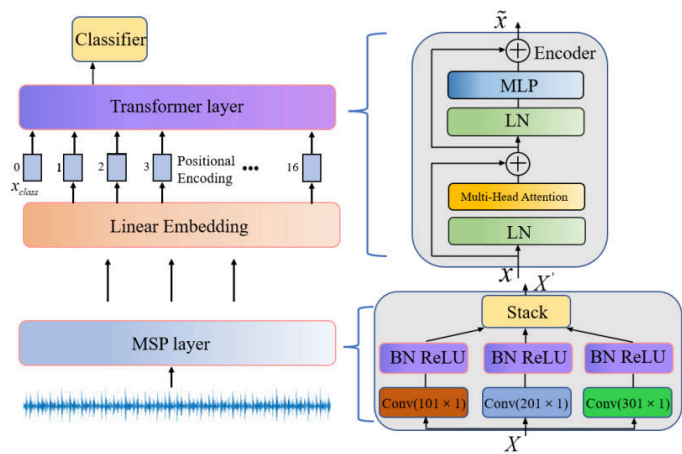


Figure 3: Improved Transformer structure diagram.

Table 1: Parameters of Improved Transformer

Number	Type	Parameters			Output
1	Input signal	-			1*1024
2	MSP layer	Kernel size	Stride	Channel	12*1024
		101/201/301	1	4	
3	Linear Embedding	Patch size	Input_dim	Output_dim	16*64*1
		64*1	768*16	64	
4	Transformer	Depth	Head	Mlp_dim	16*64*1
		6	16	128	
5	Classifier	Splicing dimensions			5
		64*1			

Table 2: Labeling rules for gearbox data

Category	Label	Health status description	
		Fault type	Working conditions
1	S1	Normal	20Hz-0V \ 30Hz-2V
2	S2	Chipped	20Hz-0V \ 30Hz-2V
3	S3	Miss	20Hz-0V \ 30Hz-2V
4	S4	Root	20Hz-0V \ 30Hz-2V
5	S5	Surface	20Hz-0V \ 30Hz-2V

4 EXPERIMENTS AND RESULTS

4.1 Dataset Description

The SEU gearbox dataset is provided by Southeast University [11] for the purpose of gearbox fault diagnosis. The dataset was collected using the Drivetrain Dynamics Simulator (DDS). Experimental device is shown in figure 4, which consists of brake, planet gearboxes, parallel gearboxes, and driving motors. In addition, two three-axis accelerometers (x, y, z) were installed on the parallel gearbox and

planet gearbox to collect vibration signals. The gearbox faults included Chipped, Miss, Crack in the root of gear feet, and wear in the surface of gear.

Two different working conditions are investigated with the rotating speed system load set to be either 20 HZ-0V or 30 HZ-2V. The dataset contains four fault labels and one normal label, making a total of five status data. Table 2 lists the fault data labeling rules at different speeds and loads.

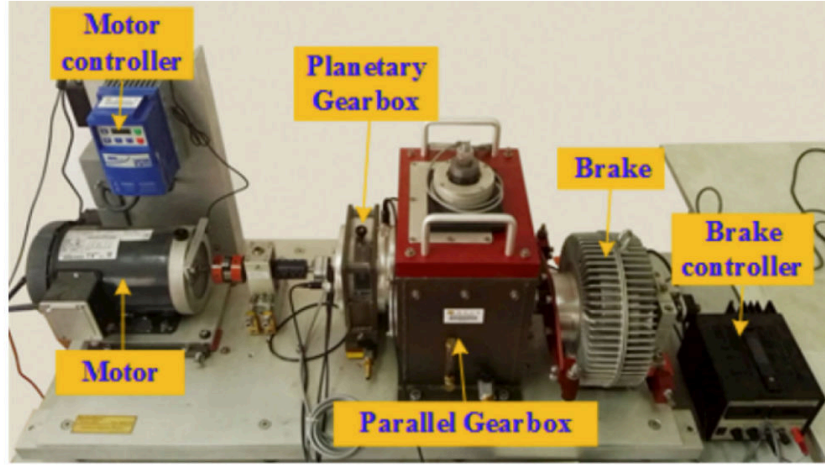


Figure 4: Experimental system for gearbox dataset.

4.2 Experiment Result and Analysis

To evaluate the fault diagnosis ability of a model on the SEU dataset, 80% of the dataset was used for training and the remaining 20% for testing. The k-fold cross-validation method was used to assist in model assessment, with k set to five to reduce random error. After approximately 50 epochs, the accuracy reached almost 99.5%. This indicates that the improved Transformer has good feature extraction ability and diagnostic accuracy.

In order to simulate the actual working conditions, noise is added to the signal to increase the difficulty of classification. Gaussian white noise based on the signal-to-noise ratio (SNR) is added. The energy relationship between signal and noise is shown in Formula (6). P_{signal} and P_{noise} represent the energy of the original signal and noise respectively. The SNR determines the amount of noise added, with SNR equal to zero indicating that the signal and noise have equal energy.

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right) \quad (6)$$

Figure 5 displays a comparison of the diagnostic capabilities of the proposed model, as well as the basic models CNN and Transformer, under conditions of noise. The model exhibits a significant improvement compared to the baseline, with higher diagnostic accuracy than the original CNN and Transformer models across various noise levels. Notably, the model can achieve an impressive diagnostic accuracy of 95% when the SNR is greater than 4, demonstrating its robustness to noise. At an SNR of 10, the model's final accuracy reaches 99.4%, indicating the effectiveness and excellence of the proposed model for fault diagnosis of gearboxes.

We utilize the T-SNE algorithm to visualize the training process in two dimensions by clustering the outputs of each layer to visualize the learning process of the model. Figure 6 shows the detailed process of network training, where different colors represent different categories. It can be observed that as the network progresses, similar features continue to gather and eventually form distinct

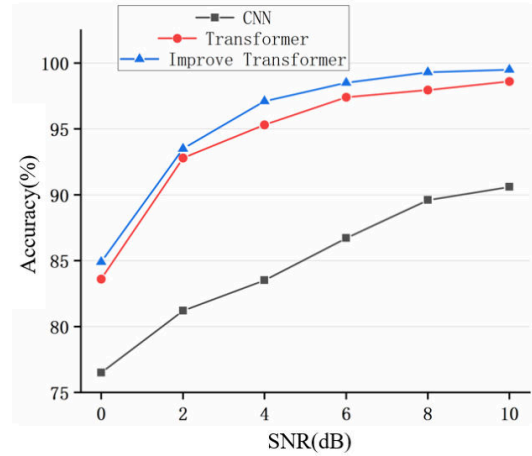


Figure 5: Diagnostic result of experimental model and baseline under different noise levels.

cluster structures. The T-SNE algorithm thus demonstrates the powerful feature learning ability of the improved model in detecting fault patterns.

Figure 7 displays the confusion matrix for each algorithm in the absence of noise. For CNN, the overall accuracy is ordinary, with some misclassifications in each class. The Transformer model exhibits low discrimination between miss teeth and wear occurring in the surface of gear, with all false positives classified as surface cracks. In contrast, the improved Transformer model exhibits the lowest error rate, indicating that the proposed model has good classification ability and has great application value in fault diagnosis.

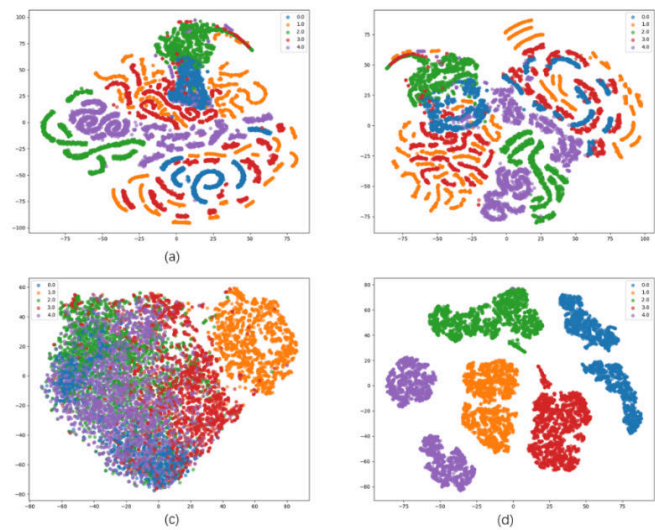


Figure 6: Display of t-SNE results for each layer of the experimental: (a) MCF, (b) Linear embed, (c) Transformer, (d) Classifier.

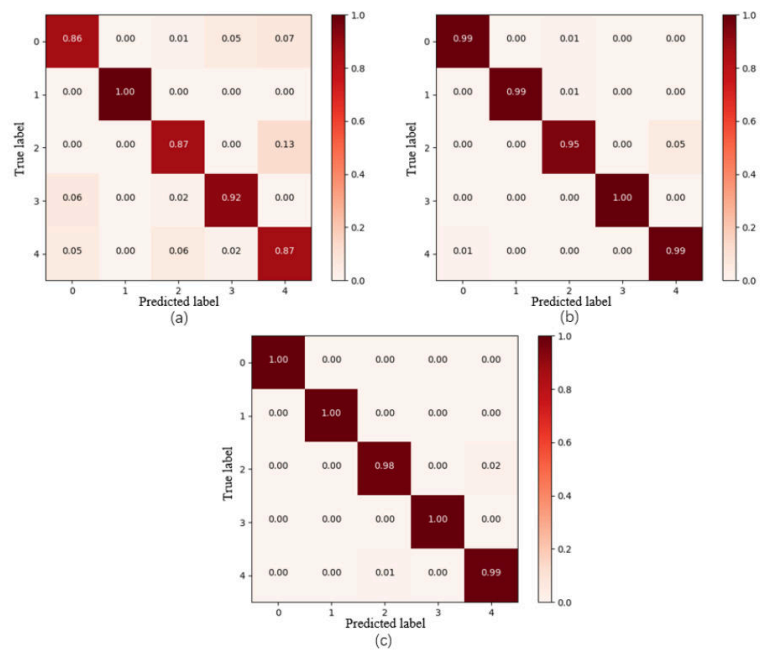


Figure 7: Confusion matrix of proposed method and baseline: (a) CNN, (b) Transformer, (c) Improved Transformer.

5 CONCLUSION

This article presents an enhanced Transformer model for intelligent fault diagnosis of gearboxes. The input signal passed through

a multi-scale perception layer, linear embedding and Transformer encoder to finally get the classification result. The SEU gearbox dataset was utilized for testing, and the results of the experiments demonstrate that the improved Transformer model is capable of

effectively capturing local feature information through the integration of multi-scale perception layers and linear embedding layers. The proposed model has the ability to analyze fault features at various time scales while retaining the position information of the original signal. During the experiment, the model outperformed other basic models in terms of diagnostic capabilities. The model's classification accuracy can reach 99.4% when detecting five different fault signals with a signal-to-noise ratio of 10. Even with a signal-to-noise ratio of 2, the model can still maintain a diagnostic accuracy of 95%. These results demonstrate the model's exceptional anti-interference ability, suggesting that the proposed method could be effective in real-world engineering fault diagnosis tasks.

In the future, we plan to use more advanced models to improve the proposed method. For example, using the recent RetNet, which has faster speed and accuracy than Transformer. We will also collect more data for different gearbox faults to evaluate the performance of the proposed method.

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sample into different regions that contain the same number of sampling points. Table 4 demonstrates the experimental results of the five methods written above.

Table 4. Accuracy of Different Models on GASF Gearbox-Based Datasets.

Model	0.72 Nm	0 Nm
TICNN	99%	99.2%
GASF+SE-ResNeXt18	99.5%	99.6%
GASF+MobileNetV3	99.7%	99.8%
MTF+TSE-ResNeXt50	99.8%	100%
GASF+TSE-ResNeXt50	100%	100%

It can be found from Table 4 that in the gearbox fault diagnosis, the TSE-ResNeXt50 model based on MTF and GASF dataset has good generalization performance. When using the GASF dataset, the accuracy of the TSE-ResNeXt50 reached 100% under two different loads. When using the MTF dataset, the accuracy of the TSE-ResNeXt50 reached 99.8% and 100% under two different loads, respectively. The accuracy of GASF+SE-ResNeXt18 and GASF+TSE-ResNeXt50 is very close, which confirms the advantages of the SE-ResNeXt module. Although the accuracy of GASF+MobileNetV3 and TICNN has also reached over 99%, overall, GASF+TSE-ResNeXt50 performs better in gear fault classification, with higher accuracy and stronger robustness.

5.4. Noise Reduction Performance Analysis

With the aim of conveniently measuring the efficiency improvement of the model after noise diminishment, Gaussian white noise with a certain signal-to-noise ratio (SNR) is added to the original vibration signal. The calculation formula of SNR is as follows

$$SNR_{dB} = 10 \log_{10}(P_{signal}/P_{noise}) \quad (20)$$

where P_{signal} is the power of the original vibration signal, and P_{noise} is the power of the added Gaussian white noise.

Figure 6 shows the comparison result between the original signal of the gearbox tooth broken under 7.32 Nm load and the noised signal with SNR = 0. As seen in the figure, after increasing the noise, the fault characteristics in the signal are interfered with to different degrees, and the fault features are not apparent. Hence, noise lessening is needed for the noisy signals to upgrade the discernment precision of the model for the fault signals containing a lot of noise.

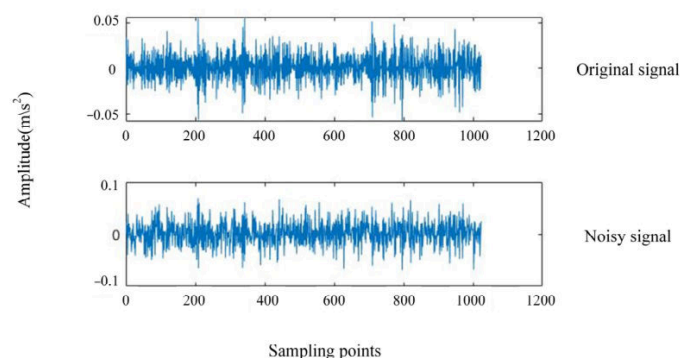


Figure 6. Comparison result between the original signal and the noised signal.

The noise reduction approach presented in this paper was adopted for noise reduction. Firstly, the ICEEMDAN algorithm was used to decompose the noised signal, and the result is presented in Figure 7. As can be found from the figure, after ICEEMDAN decomposition,

characteristics, more significant vibration and shock, and a higher coincidence degree with the original signal compared with the signal without noise reduction.

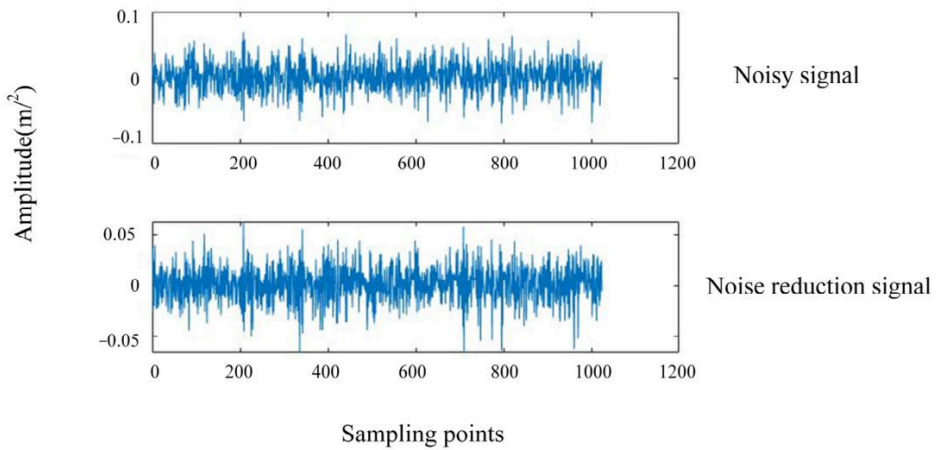


Figure 8. The comparison result between the denoised signal and the noised signal.

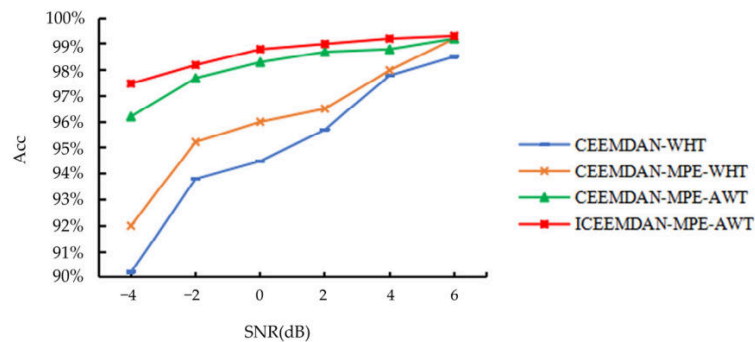
In order to verify the efficacy of the ICEEMDAN-MPE-AWT noise reduction method in improving the fault diagnosis of the model in noisy environments, the experimental dataset with a load of 7.32 Nm was selected, and noise with SNR ranging from -4 dB to 6 dB was introduced to the test set. Then, the precision of the raised method before and after noise reduction was measured and compared to prove the effectiveness of the raised approach. For the sake of checking the generalization of the noise reduction method raised in this paper, the GASF+TSE-ResNeXt50 model and GASF+TSE-ResNeXt18 were selected as the test model for fault diagnosis. The recognition accuracy of the model on the test set was taken as the evaluation index, 10 repeated tests were carried out for each model, and the average value of the 10 tests was seen as the final results. The results are illustrated in Table 6. It can be found from Table 6 that after noise reduction by adopting the approach in this paper, the accuracy of GASF+TSE-ResNeXt50 can still reach 98.8% when the SNR is -4 dB. Therefore, the approach introduced in this paper has better noise reduction efficacy in noisy environments and can ensure the precision of the model. Figure 9 shows the precision curves of the two models before and after noise reduction, which can more intuitively see the effectiveness of ICEEMDAN-MPE-AWT noise reduction method. Figure 10 shows the confusion matrices of GASF+TSE-ResNeXt50 before and after denoising when SNR is -4 dB and the load is 7.32 Nm. The horizontal coordinate represents the true label of the failure, the vertical coordinate represents the predicted category of the failure, and the numbers on the main diagonal of the matrix represent the number of samples correctly classified for each type of failure.

Table 6. Accuracy under different SNR.

Model	SNR (dB)	−4	−2	0	2	4	6
GASF+TSE-ResNeXt50	Before noise reduction	95.5%	97.5%	98.5%	99.0%	99.7%	100%
	After noise reduction	98.8%	99.2%	99.7%	99.8%	100%	100%
GASF+TSE-ResNeXt18	Before noise reduction	92.5%	94.2%	96.3%	97.7%	98.2%	99.2%
	After noise reduction	97.5%	98.2%	98.8%	99%	99.2%	99.3%

Table 7. Accuracy of each denoising method on GASF+ TSE-ResNeXt18.

SNR (dB)	−4	−2	0	2	4	6
CEEMDAN-WHT	90.2%	93.8%	94.5%	95.7%	97.8%	98.5%
CEEMDAN-MPE-WHT	92.0%	95.2%	96.0%	96.5%	98.0%	99.2%
CEEMDAN-MPE-AWT	96.2%	97.7%	98.3%	98.7%	98.8%	99.2%
ICEEMDAN-MPE-AWT	97.5%	98.2%	98.8%	99.0%	99.2%	99.3%

**Figure 11.** Each denoising method on GASF+SE ResNeXt18.

Compared with CEEMDAN-MPE-AWT, the accuracy of CEEMDAN-MPE-WHT decreased significantly; this indicates that setting wavelet coefficients below the threshold to zero in WHT may give rise to the loss of elaborate information in the signal. Especially for some weak detail signals or low amplitude variation signals, WHT may not be able to effectively retain this detailed information. In addition, WHT needs to manually select the appropriate threshold for signal denoising. However, in practical applications, especially when the statistical characteristics of the signal are not clear or the noise level changes greatly, it is often very difficult to determine a suitable threshold value. Therefore, the accuracy of CEEMDAN-MPE-WHT has decreased to some extent. The AWT used in this paper can adjust the threshold dynamically by introducing a correction factor, and the calculation results of the threshold can be further adjusted by introducing a convergence factor $\theta = k/L$, thereby improving the resolution of low-frequency signals and enhancing the denoising effect. This method can dynamically adjust the threshold value according to the local characteristics of the signal, which can better keep the details and edge information of the signal, has strong adaptability, and improves the stability and accuracy of denoising.

Compared with CEEMDAN-MPE-WHT, CEEMDAN-WHT has similar accuracy when the SNR is high. However, when the SNR decreases, CEEMDAN-WHT fluctuates significantly. This indicates that the cross-correlation coefficient may not be able to accurately distinguish the noise component from the signal component, and it may be easy to divide the low-frequency component into noise, resulting in certain fluctuations in accuracy. In contrast, MPE can reveal the multiscale structure of a signal by calculating the arrangement entropy at different scales, which allows it to consider both local and global characteristics of the signal, thus, providing more comprehensive signal analysis results. At the same time, MPE is robust to noise and interference. It can measure the complexity and randomness of the signal by calculating the arrangement entropy of the signal so that the interference of noise can be reduced to a certain extent. Therefore, CEEMDAN-MPE-WHT has relatively good fault diagnosis accuracy. In summary, ICEEMDAN-MPE-AWT proposed in this paper has better noise reduction performance than other methods in noisy circumstances and can improve the accuracy of the model for noisy signals.

6. Conclusions

This paper focuses on the problem of the signal collected by sensors including a great deal of noise in complex working environments, making the model unable to effectively

body failure, inner-ring–outer-ring mixed failure, normal condition, inner-ring failure, and outer-ring failure, as shown in Table 3.

Table 3. Data set of rolling bearings of SEU.

Operation Conditions	Dataset	Data Length	Fault Type
I	Ball_20_0	1,048,560	Rolling-body failure
	Comb_20_0	1,048,560	Inner-ring and outer-ring mixed failure
	Health_20_0	1,048,560	Normal condition
	Inner_20_0	1,048,560	Inner-ring failure
	Outer_20_0	1,048,560	Outer-ring failure
II	Ball_30_2	1,048,560	Rolling-body failure
	Comb_30_2	1,048,560	Inner-ring and outer-ring mixed failure
	Health_30_2	1,048,560	Normal condition
	Inner_30_2	1,048,560	Inner-ring failure
	Outer_30_2	1,048,560	Outer-ring failure

In the planetary-gear dataset, there were five types of broken-tooth faults: normal conditions, missing-tooth faults, root faults, and tooth-surface faults, as shown in Table 4.

Table 4. Planetary-gear dataset of SEU.

Operation Conditions	Dataset	Data Length	Fault Type
I	Chipped_20_0	1,048,560	Broken-tooth failure
	Health_20_0	1,048,560	Normal condition
	Miss_20_0	1,048,560	Missing-tooth failure
	Root_20_0	1,048,560	Tooth-root failure
	Surface_20_0	1,048,560	Tooth-surface failure
II	Chipped_30_2	1,048,560	Broken-tooth failure
	Health_30_2	1,048,560	Normal condition
	Miss_30_2	1,048,560	Missing-tooth failure
	Root_30_2	1,048,560	Tooth-root failure
	Surface_30_2	1,048,560	Tooth-surface failure

4.2. Analytical Comparison

The energy entropy and fuzzy entropy of each component were obtained by decomposing the bearing- and gear-vibration signals through VMD and SVMD. Each of them was constructed into an energy-entropy sample set and a fuzzy-entropy sample set. A machine-learning [32,33] classification model was established to perform intelligent diagnosis of faults.

Firstly, the gear data of two working conditions of the SEU dataset were be studied, and the data length of each class of gear was 1,048,560. This paper took the length of 4096 as a data sample, which could be divided into 255 in total, and then used the VMD and SVMD methods for mode decomposition of each data sample.

When computing energy entropy and fuzzy entropy as feature vectors, it is generally sufficient to take the first six components. The disadvantage of choosing more or fewer components is that choosing more components can increase the computation burden and may lead to overfitting problems, where the model performs well on training data but poorly on unknown data. Additionally, too many components can increase the complexity of the model, making it harder to interpret and understand. Choosing fewer components may result in the loss of important information and a decrease in model accuracy. Therefore, selecting an appropriate number of components, specifically the first six orders, can ensure model accuracy while avoiding issues of excessive complexity and computational burden. Therefore, in this paper, we sought the energy-entropy value and fuzzy-entropy value

methods, such as boosting in AdaBoost and gradient-boost decision tree (GBDT) [36]. The other is parallel integration methods, such as bagging and random forest (RF) [37].

The classification results of the energy-entropy and fuzzy-entropy sample sets obtained from the VMD and SVMD decompositions of the bearing data in the gearbox of SEU in the SVM, k-NN, RF, and GBDT models are shown in Table 5. The table shows that the accuracy of the SVMD fuzzy entropy k-NN, RF, and GBDT models in the gearbox working condition I data of SEU was the same as that of the VMD fuzzy-entropy RF and GBDT models, reaching 99.22%; the accuracy of the VMD fuzzy-entropy SVM and RF models in the working condition II data was the highest, reaching 99.61%.

Table 5. Accuracy of gearbox bearing data at SEU.

Data		Gearbox Bearing Data of SEU							
Operation Condition		Operation Condition I				Operation Condition II			
Model	SVM	k-NN	RF	GBDT	SVM	k-NN	RF	GBDT	
VMD energy entropy	97.57%	97.34%	98.04%	98.04%	97.25%	96.86%	96.08%	97.25%	
SVMD energy entropy	97.25%	96.86%	96.47%	97.25%	94.12%	93.33%	91.76%	94.51%	
VMD fuzzy entropy	99.13%	98.98%	99.22%	99.22%	99.61%	92.16%	99.61%	99.22%	
SVMD fuzzy entropy	93.96%	99.22%	99.22%	99.22%	94.90%	97.25%	95.69%	95.29%	

The SVMD feature-extraction (energy entropy, fuzzy entropy) machine learning in this paper was compared with the machine-learning classification results in the literature [38], as shown in Table 6. From the table, it can be seen that in the fault diagnosis of the bearing, the machine-learning classification effect of the SVMD method was better than the identification result of this data in the literature, and in the gearbox, the two were difficult to distinguish.

Table 6. Comparison of SVMD feature extraction and classification results and those from other literature.

Algorithm		Component: Bearing		Component: Gear	
		20-0	30-2	20-0	30-2
Literature [36]	K-NN	80.80%	86.40%	93.20%	89.20%
	SVM	83.30%	88.60%	94.40%	90.10%
Our paper	SVMD fuzzy entropy + k-NN	99.22%	97.25%	94.42%	90.98%
	SVMD energy entropy + SVM	97.25%	94.12%	91.27%	90.20%
	SVMD fuzzy entropy + RF	99.22%	95.69%	96.02%	93.33%
	SVMD fuzzy entropy + GBDT	99.22%	95.29%	95.22%	89.41%

In the paper, the grid-search method was also chosen to select the two model parameters of RF and GBDT. When performing the grid search, the n_estimators parameter and the max_depth parameter in RF were both set to integers between 5 and 13, the min_samples_split parameter was set to 2, and other parameters were set by default. In GBDT, the n_estimators parameter was set to an integer between 50 and 150 with an interval of 10, and other parameters were set by default.

4.3. Additional Dataset Validation

In addition to Southeast University experimental dataset introduced above, to further validate the proposed method, we also used the experimental data collected by the GDS3000 (SpectraQuest, Inc., Richmond, VA, USA) experimental platform. The platform is equipped with switchable normal bearings, inner fault bearings, outer fault bearings, rolling-element fault bearings, and compound fault bearings (with inner and outer fault), as shown in Figure 14.

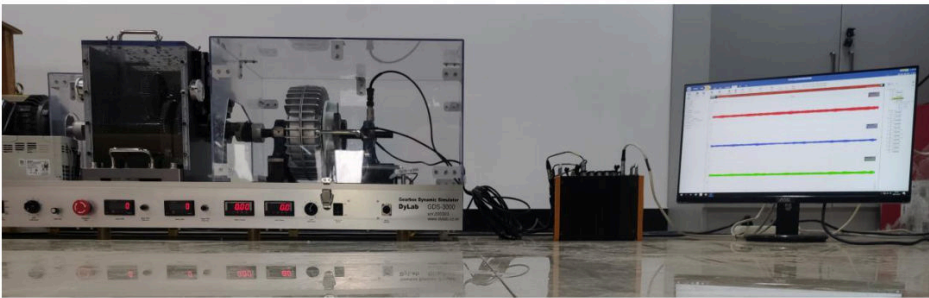


Figure 14. GDS3000 experimental platform.

Using the GDS3000 experimental platform, data were collected based on two working conditions: medium–high speed (1540 r/min) and high speed (1788 r/min). The experimental results in Table 7 show that the fuzzy entropy was superior to the energy entropy.

Table 7. Results of GDS 3000 experimental platform.

Operation Conditions	Medium–High Speed (1540 r/min)				High Speed (1788 r/min)			
	SVM	KNN	RF	GBDT	SVM	KNN	RF	GBDT
VMD energy entropy	98.40%	98.53%	98.53%	97.44%	98.53%	98.53%	98.17%	97.44%
SVMD energy entropy	98.17%	95.60%	95.24%	94.14%	94.46%	97.42%	94.46%	94.10%
VMD fuzzy entropy	94.56%	98.40%	98.53%	98.17%	96.70%	98.90%	91.58%	95.97%
SVMD fuzzy entropy	97.07%	98.53%	95.97%	96.34%	96.31%	97.42%	95.94%	94.84%

5. Discussion and Open Issues

5.1. Classification-Model Design

In this paper, we used four classification models: SVM, KNN, RF, and GBDT. Table 8 shows the comparison of the classification models for reference. However, in practical industrial applications, the design of a classification method requires the consideration of several factors, including the type of data, the type of classification problem, the size of the data set and the number of features, the accuracy requirements, the interpretability, and the complexity of the model. When designing a model, these factors need to be considered comprehensively, and the most appropriate model should be selected based on practical applications.

Table 8. Comparison of classification models.

Model	Applicable Data	Classification Problem	Training Time	Storage Space
SVM	Low-dimensional	Linear classification	Long	Low
KNN	Low-dimensional	Nonlinear classification	Short	High
RF	High-dimensional	Nonlinear classification	Long	Low
GBDT	High-dimensional	Nonlinear classification	Long	Low

5.2. Research Limitations

In terms of the limitations of our research, we agree that further optimization of the classification-model parameters is needed to improve accuracy. Additionally, our study focused on a limited dataset, and more industrial data are needed to validate the model’s performance in real-world applications.

As for future research directions, we plan to explore more advanced optimization techniques to further improve the accuracy of the classification model. We also plan to collect more diverse and comprehensive industrial data to better validate the model’s performance in various industrial applications. In addition, we plan to investigate the potential of incorporating other datasets to further enhance the model’s accuracy. We

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Table 2
Description of PU dataset (Case 1)

Category	Bearing	Damage	Level	Method	Label
0	K001	Normal	/	/	H_1
1	K002	Normal	/	/	H_2
2	K003	Normal	/	/	H_3
3	K004	Normal	/	/	H_4
4	K005	Normal	/	/	H_5
5	K006	Normal	/	/	H_6
6	KA01	Outer ring	1	EDM damage	H_7
7	KA03	Outer ring	2	electric engraver	H_8
8	KA05	Outer ring	1	EDM damage	H_9
9	KA06	Outer ring	2	electric engraver	H_{10}
10	KA07	Outer ring	1	drilling	H_{11}
11	KA08	Outer ring	2	drilling	H_{12}
12	KA09	Outer ring	2	drilling	H_{13}
13	KI01	Inner ring	1	EDM damage	H_{14}
14	KI03	Inner ring	1	electric engraver	H_{15}
15	KI05	Inner ring	1	electric engraver	H_{16}
16	KI07	Inner ring	2	electric engraver	H_{17}
17	KI08	Inner ring	2	electric engraver	H_{18}

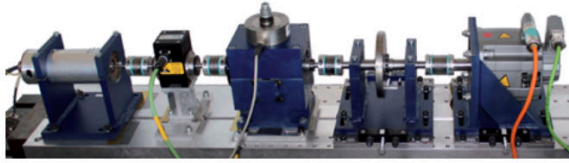


Fig. 5: Rolling bearing test rig. (Case 1)

experiment, the SNR is set in the range from -10 dB to -2 dB. Each group of experiments was repeated five times, with 200 epochs assigned to each experiment. Table 1 shows the network structure of FE-MCFormer-s and FE-MCFormer, where FE-MCFormer-s is the lightweight version of the proposed method with fewer parameters and FE-MCFormer is the standard version with standard parameters. In the l -th MSTFF block, $B1$ and $B2$ represent convolution kernels of size 1×5 and 1×3 in Eq. (8), respectively; $\gamma = 0.1$ denotes the scaling factor γ in Eq. (5) and Eq. (14) is set to 0.1; C represents the number of classes in each dataset.

3.2. Case 1: Rolling Bearing Dataset

The bearing dataset [23] from Paderborn University (PU) is employed to validate the FE-MCFormer under strong noise conditions.

3.2.1. Dataset description

As shown in Fig. 5, the rolling bearing test bench comprises an induction motor, a load motor, a measurement shaft, and a control panel. A deep groove ball bearing is selected as the test object. The experiment is conducted with the induction motor operating at 1500 rpm under a torque load of 0.7 Nm, with a sampling frequency of 64 kHz. A total of 18 bearing states are employed, including 6 normal

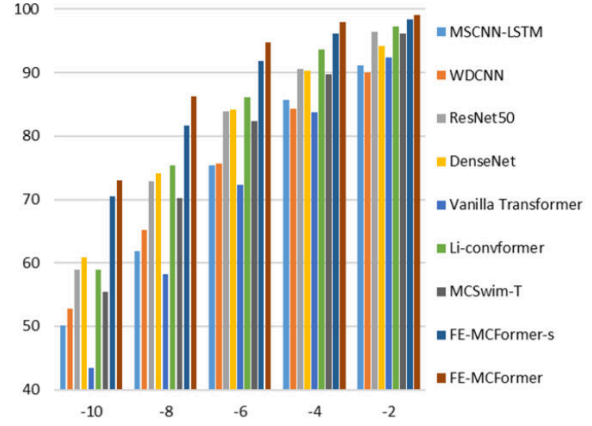


Fig. 6: Test results of rolling bearing dataset in Case 1.

states, 7 outer-ring faults, and 5 inner-ring faults, as detailed in Table 2. For each health state, the raw vibration signal was segmented sequentially into 1000 consecutive samples, resulting in a total of $18 \times 1000 = 18,000$ samples, with each sample containing 2048 vibration points. To avoid data leakage, the first 80% of the sequentially segmented samples were used for model training, while the remaining 20% were reserved for testing.

3.2.2. Overall Diagnostic Performance

To assess the effectiveness of FE-MCFormer, extensive experiments are conducted under varying noise conditions, with SNR ranging from -10 dB to -2 dB. Seven approaches, namely MSCNN-LSTM [24], Vanilla Transformer [16], WDCNN, ResNet50 [25], DenseNet, MCSwin-T [26] and Li-convformer [27], are employed for comparison with FE-MCFormer. From Table 3 and Fig. 6, it can be observed that the proposed method consistently outperforms existing data-driven baselines across all noise levels in terms of diagnostic performance.

Specifically, FE-MCFormer and FE-MCFormer-s achieve superior performance across all noise scenarios, significantly outperforming both CNN-based and transformer-based methods. The proposed network architecture maintains high accuracy even under extremely noisy environments, further illustrating its strong robustness. Fig. 7 shows the 2D T-SNE visualization results of various methods under SNR = -6 dB, the results show dense clustering of different fault types with high classification accuracy, which further illustrates the feature discrimination capability of FE-MCFormer in complex industrial environments. These results indicate that FE-MCFormer architecture maintains a robust feature extraction capability even under severe noise interference, which is critical for industrial condition monitoring scenarios.

3.2.3. Comparative Analysis

MSCNN-LSTM, as a state-of-the-art multiscale convolutional network, exhibits outstanding performance under

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Table 3
Test results of various data-driven methods on PU dataset in Case 1 (%)

Methods	SNR=-10	SNR=-8	SNR=-6	SNR=-4	SNR=-2	Params(M)	FLOPs(M)
	Acc	Acc	Acc	Acc	Acc		
MSCNN-LSTM	50.17±2.17	61.90±5.66	75.39±5.09	85.65±1.11	91.13±1.20	0.09	57.37
WDCNN	52.81±1.53	65.23±0.72	75.70±1.51	84.23±1.12	90.01±0.67	0.04	0.65
ResNet50	58.85±0.47	72.90±0.62	83.86±0.54	90.57±0.47	96.49±0.34	15.99	1610.34
DenseNet	60.81±0.33	74.11±0.72	84.15±0.39	90.31±0.92	94.15±0.59	3.80	526.17
Vanilla Transformer	43.43±2.16	58.20±0.43	72.29±0.25	83.72±0.21	92.43±0.58	3.19	266.95
Li-convformer	58.98±0.34	75.34±0.68	86.12±0.32	93.65±0.33	97.29±0.38	0.32	28.83
MCSwin-T	55.49±0.46	70.16±0.10	82.34±0.26	89.74±0.19	96.14±0.25	1.94	453.66
FE-MCFormer-s	70.44±1.26	81.70±1.28	91.90±0.64	96.16±0.44	98.42±0.15	2.43	54.60
FE-MCFormer	72.99±0.54	86.31±0.51	94.74±0.17	98.01±0.14	99.11±0.18	7.24	338.95

weak noise conditions, reaching 91.13% mean accuracy at the SNR of -2 dB. However, as the noise level increases from -2 dB to -6 dB, its accuracy drops significantly. In contrast, standard FE-MCFormer achieves diagnostic accuracy of 99.11%, 98.01%, and 94.74% under the SNR from -2 dB to -6 dB. Numerical results indicate that, in noisy environments, the multiscale kernels in the multiscale convolutional neural networks (MSCNN) tend to amplify both high-frequency noise patterns and key local features, resulting in redundant and conflicting weight distributions. This makes it difficult for the network to focus on critical fault features, resulting in degraded performance in the FD of rotating machines. In contrast, the proposed FE-MCFormer leverages FALL and TFFN to suppress irrelevant frequency components and employs MSCAL to strengthen local multiscale feature representations, thereby achieving superior performance over MSCNNs.

The WDCNN uses large kernels to capture global dependencies, thus reducing noise interference. However, due to the limitation of the convolution kernel, the model struggles to fully capture long-range dependencies over the entire signal sequence. More specifically, as illustrated in Table 3, the mean accuracies of WDCNN under SNR = -2 , -4 , -6 , -8 , -10 are 90.01%, 84.23%, 75.70%, 65.23% and 52.81%. Compared with standard FE-MCFormer and FE-MCFormer-s, FE-MCFormer consistently achieves higher accuracy under all noise conditions, indicating the effectiveness of FE-MCFormer.

Compared with ResNet50 and DenseNet, FE-MCFormer achieves higher diagnostic accuracy under all noise conditions. This improvement can be attributed to its time–frequency modeling strategy. In conventional CNN architectures such as ResNet50 and DenseNet, small convolutional kernels mainly capture local patterns within a limited receptive field, and their responses can be easily affected by noise-induced fluctuations under low-SNR conditions. In contrast, FE-MCFormer combines frequency adaptive learning with multiscale temporal feature extraction, enabling the model to suppress noise-dominated components while retaining

fault-sensitive structures. As reported in Table 3, the mean accuracies of the standard FE-MCFormer under SNR levels of -2 , -4 , -6 , -8 , and -10 dB are 99.11%, 98.01%, 94.74%, 86.31%, and 72.99%, respectively. These results are consistently higher than those of ResNet50, which obtains 96.49%, 90.57%, 83.86%, 72.90%, and 58.85%, respectively. Compared with DenseNet, FE-MCFormer also achieves accuracy improvements of 4.96%, 7.70%, 10.59%, 12.20%, and 12.18% across the same noise levels. It is also worth noting that FE-MCFormer maintains lower computational cost than these CNN baselines. Specifically, the computational costs of FE-MCFormer and FE-MCFormer-s are 338.95M and 54.60M FLOPs, respectively, which are lower than those of ResNet50 (1610.34M FLOPs) and DenseNet (526.17M FLOPs). These results indicate that the proposed framework achieves a favorable balance between noise-robust diagnostic performance and computational efficiency compared with representative CNN-based models.

Vanilla Transformer shows a dramatic decrease in diagnostic performance under strong noise. This is mainly attributed to the fact that the severe noise contaminates the entire signal sequence, which prevents the MHSA from effectively modeling global features. The proposed FE-MCFormer utilizes FALL and TFFN to suppress irrelevant information and capture global dependencies. The MSCAL and distillation mechanism are further introduced to precisely extract the key local features, thus achieving higher performance.

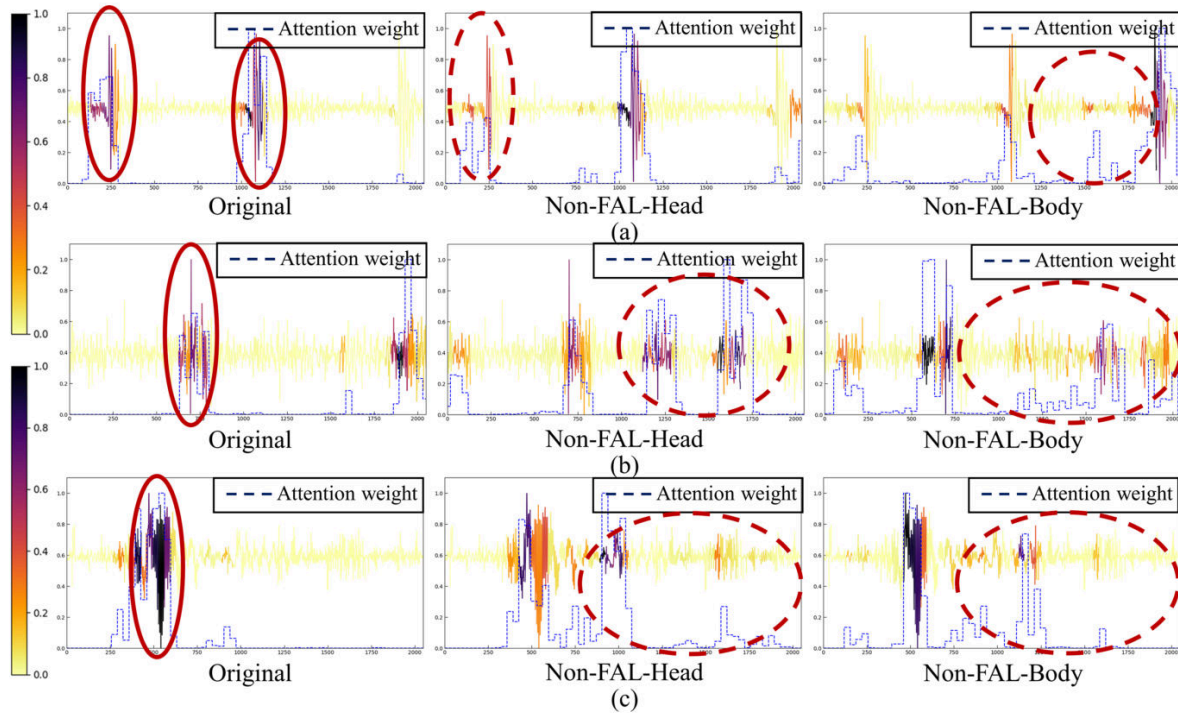
Li-convformer, as a state-of-the-art lightweight convformer-based fault diagnosis model, achieves competitive performance under relatively weak noise conditions, with mean accuracies of 97.29%, 93.65%, and 86.12% at SNR levels of -2 , -4 , and -6 dB, respectively. However, its diagnostic performance degrades noticeably as the noise intensity increases. In comparison, FE-MCFormer-s obtains mean accuracies of 98.42%, 96.16%, and 91.90% under the same SNR levels, indicating improved robustness under noisy conditions. Moreover, FE-MCFormer-s maintains stable

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Table 4

Results of ablation study on PU dataset in Case 1

Methods	SNR=-10	SNR=-8	SNR=-6	SNR=-4	SNR=-2	Params(M)	FLOPs(M)
	Acc	Acc	Acc	Acc	Acc		
FE-MCFormer	72.99±0.54	86.31±0.51	94.74±0.17	98.01±0.14	99.11±0.18	7.24	338.95
Non-MSA	70.78±1.09	82.59±1.34	91.61±0.75	95.80±0.51	98.08±0.31	7.24	263.50
Non-FAL-Head	69.77±0.81	81.28±0.33	90.99±0.74	95.95±0.58	98.22±0.40	6.98	338.07
Non-FAL-Body	70.01±0.74	83.40±0.86	91.83±0.59	96.51±0.27	98.54±0.97	7.11	271.71

**Fig. 12:** Attention heat map of FE-MCFormer under SNR = -8 dB: (a) KA01, outer ring. (b) KA07, outer ring. (c) KI01, inner ring.

reconstructed signals in test dataset produced by FALL are visualized in both the frequency domain and envelope spectrum. For illustration, Fig. 8, Fig. 9, Fig. 10, and Fig. 11 present the reconstruction results under SNR = -4 dB, where the blue curve denotes the noisy signal, the orange curve represents the FALL-reconstructed signal, and the pink shaded bands indicate the frequency-resolution regions around the ball pass frequency of outer race (BPFO) or the ball pass frequency of inner race (BPFI) and their harmonics.

As shown in Fig. 8, Fig. 9, Fig. 10, and Fig. 11, the reconstructed spectra suppress part of the broadband high-frequency noise components while retaining several fault-sensitive spectral responses. More importantly, the envelope spectra of the reconstructed signals exhibit pronounced responses within the frequency-resolution bands around the

theoretical fault characteristic frequencies and their harmonics. Specifically, for the outer-race faults KA01 and KA03 shown in Fig. 8 and Fig. 9, the dominant responses are distributed around the BPFO-related bands, including approximately 76 Hz, 152 Hz, 228 Hz, 305 Hz, and their neighboring harmonic regions. For the inner-race faults KI01 and KI07 shown in Fig. 10 and Fig. 11, clear responses can be observed around the BPFI-related bands, including approximately 123 Hz, 247 Hz, 371 Hz, 494 Hz, and their neighboring harmonic regions. These results indicate that FALL can effectively suppress noise-dominated spectral components while preserving physically meaningful fault-related periodic modulation features, thereby providing frequency-domain interpretability for the proposed model.

To illustrate the interpretability of the proposed method in the time domain, we visualize the attention heatmaps

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Table 6

Test results on the compressor dataset in Case 2 (%)

Methods	SNR=-10	SNR=-8	SNR=-6	SNR=-4	SNR=-2	Params(M)	FLOPs(M)
	Acc	Acc	Acc	Acc	Acc		
MSCNN-LSTM	63.69±2.04	72.05±2.29	74.81±1.75	88.70±0.63	90.54±2.17	0.092	57.37
WDCNN	74.52±2.94	86.04±0.86	91.53±1.39	94.12±0.33	96.84±0.20	0.042	0.65
ResNet50	75.90±0.87	84.01±0.49	89.90±0.37	94.05±0.22	94.99±0.50	15.99	1610.34
DenseNet	81.56±1.22	88.39±0.41	93.00±0.49	95.32±0.44	96.66±0.61	3.80	526.17
Vanilla Transformer	62.70±0.49	72.72±0.91	81.17±0.40	87.49±0.44	92.76±0.51	3.19	266.95
Li-convformer	89.51±0.25	93.24±0.39	95.73±0.48	97.62±0.25	98.28±0.27	0.32	28.83
MCSwin-T	63.90±0.61	73.78±1.53	84.16±1.51	89.52±0.79	94.71±0.63	1.94	453.66
FE-MCFormer-s	93.25±0.30	96.97±0.14	98.35±0.17	99.61±0.08	99.93±0.04	2.43	54.60
FE-MCFormer	93.48±1.18	96.63±0.15	98.46±0.18	99.71±0.07	99.98±0.04	7.24	338.95

Meanwhile, the attention distributions learned by MSA become increasingly dispersed and susceptible to noise interference, resulting in reduced feature discrimination capability and poor interpretability. In contrast, the complete FE-MCFormer architecture exhibits highly concentrated attention around fault-related impulsive regions. This phenomenon suggests that FAL suppresses irrelevant spectral components and preserves fault-sensitive harmonic structures. As a result, MSA can focus more consistently on diagnostically informative temporal regions, thereby enhancing both the noise robustness and interpretability of FE-MCFormer under severe noise conditions.

3.3. Case 2: Centrifugal Compressor Dataset

3.3.1. Dataset description

On the basis of fault diagnosis studies on small-scale rotating machinery, this work further extends the investigation to fault identification of large-scale turbomachinery. Large-scale compressors are critical turbomachines in the energy, petrochemical, metallurgical, and manufacturing industries, and their operating conditions are directly associated with production continuity, equipment safety, and energy utilization efficiency. Once faults occur, they may lead to increased vibration, degraded operating efficiency, and unexpected shutdowns, resulting in substantial economic losses and safety risks. Moreover, large-scale compressors usually operate in complex industrial environments, where the collected signals are easily affected by background noise, which poses a challenge to the robustness of diagnostic models. To evaluate the practical applicability of FE-MCFormer, a real-world centrifugal compressor dataset is employed. The 7200 and 3300 eddy current probes are utilized to acquire the bearing displacement signals; the sampling frequency is set to 1024 Hz. In this study, the shaft displacement signals in the x-direction of the compressor bearing are collected under five operating conditions, including four fault states and one normal state, with 1000 samples for each condition, as summarized in Table 5. Representative fault modes of the centrifugal compressor unit are illustrated in Fig. 13, such as

block friction and blade rupture. Meanwhile, similar to the experiment in the PU dataset, the first 80% of samples are used for model training, and the remaining 20% are used for testing. Each sample contains 2048 points. All samples are normalized according to Eq. (24).

3.3.2. Comparative Analysis

In this case, Gaussian white noise with SNR levels ranging from -10 dB to -2 dB is added to the raw signals to evaluate the robustness of the proposed method. Table 6 reports the diagnostic performance of various methods in this case. As shown in Table 6, the proposed FE-MCFormer family achieves the best or highly competitive performance among the compared methods under all noisy conditions. More specifically, compared with MSCNN-LSTM and WDCNN, the proposed method provides clear accuracy improvements, particularly at low SNRs, indicating that frequency adaptive learning is beneficial for suppressing noise-dominated responses in compressor signals. Compared with ResNet50 and DenseNet, FE-MCFormer obtains higher accuracy with lower computational cost, while FE-MCFormer-s further reduces the FLOPs to 54.60M and still maintains competitive accuracy. In comparison with Transformer-based models such as the Vanilla Transformer and MCSwin-T, FE-MCFormer shows better robustness and avoids the high computational burden introduced by attention-based architectures. Moreover, compared with the strong lightweight convformer baseline Li-convformer, FE-MCFormer-s achieves higher accuracies across all SNR levels, with gains ranging from 1.65% to 3.74%. These results indicate that the proposed framework provides an effective trade-off between noise-robust fault diagnosis performance and computational efficiency for centrifugal compressor applications.

3.3.3. Interpretability Analysis

It is worth noting that, unlike rolling-bearing faults, the abnormal trajectory of the shaft centerline caused by compressor failure does not result in obvious temporal impulsive

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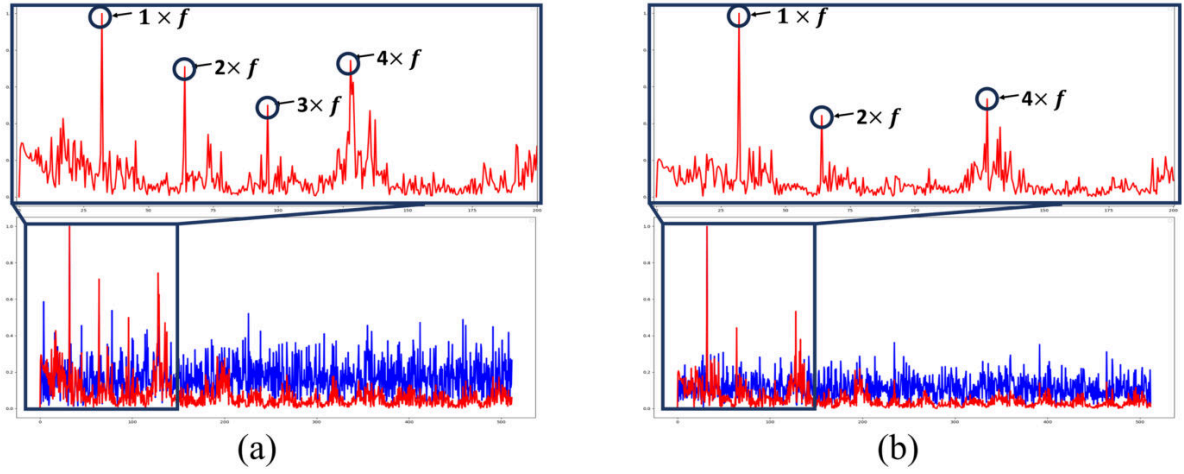


Fig. 14: Reconstruction results of the compressor dataset with four different types of faults described in Table 5 (Case 2) under $\text{SNR} = -8$ dB, where the red curve denotes the reconstruction results, the blue curve denotes the original noisy signal. (a) Blade rupture. (b) Block friction.

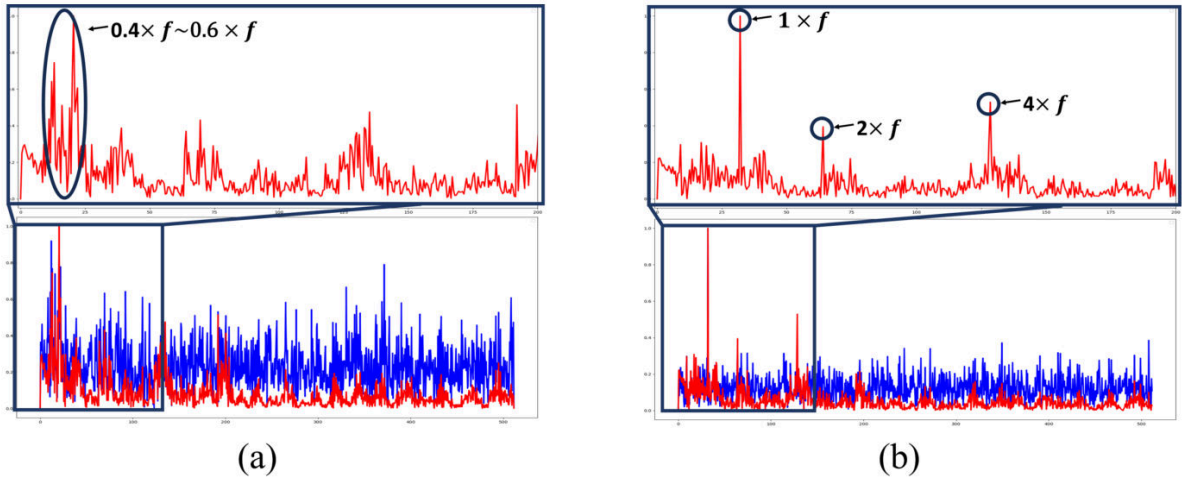


Fig. 15: Reconstruction results of the compressor dataset with four different types of faults described in Table 5 (Case 2) under $\text{SNR} = -8$ dB, where the red curve denotes the reconstruction results, the blue curve denotes the original noisy signal. (a) Oil whirl. (b) Rotor imbalance.

patterns in the time domain. Therefore, this section mainly focuses on analyzing the interpretability of the model in the frequency domain. To demonstrate the practical interpretability and effectiveness of FE-MCFormer, we compare the normalized frequency spectra of the original and reconstructed signals of FALL in test dataset under $\text{SNR} = -8$ dB, as illustrated in Fig. 14 and Fig. 15, where the blue curves denote the original spectra, and the red ones correspond to the reconstructed signals; The horizontal axis represents frequency, and the vertical axis represents amplitude. Here, $i \times f$ denotes the fault rotational frequency and its harmonics.

It can be observed from Fig. 14 and Fig. 15 that FALL attenuates substantial high-frequency noise while preserving fault-sensitive spectral components and their harmonics. For

instance, in the compressor blade rupture case shown in Fig. 14(a), the signal reconstructed by FALL retains the dominant low-frequency components associated with the fault condition, including the rotational component at 32 Hz ($1 \times f$) and its harmonics at 64 Hz ($2 \times f$), 96 Hz ($3 \times f$), and 128 Hz ($4 \times f$).

For the block-friction case shown in Fig. 14(b), pronounced spectral peaks can also be observed at the rotational frequency of 32 Hz ($1 \times f$) and its harmonic components, such as 64 Hz ($2 \times f$) and 128 Hz ($4 \times f$). This is consistent with the nonlinear vibration characteristics commonly induced by rubbing-related faults. In contrast, for the oil-whirl case shown in Fig. 15(a), particular attention should be paid to the preservation of sub-synchronous components

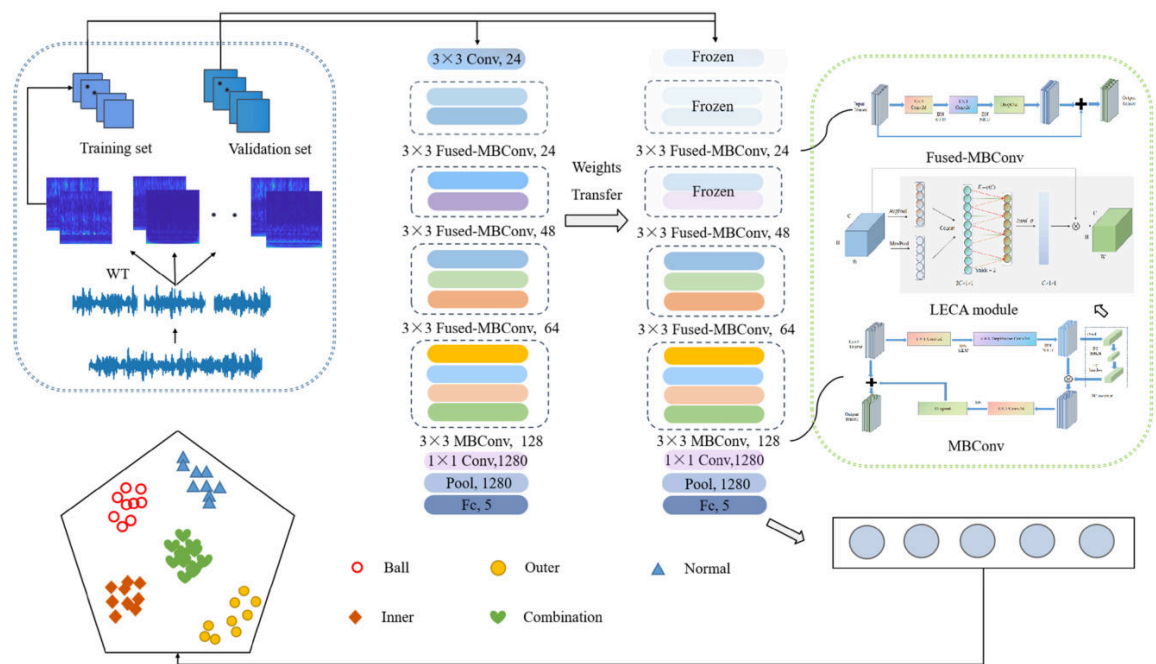


Figure 5. Fault diagnosis flow of the LECA-EfficientNetV2 transfer learning model.

1. Sample pretreatment. Wavelet transform is performed on original signals to obtain RGB images ($224 \times 224 \times 3$). All the obtained samples are divided into training sets and validation sets, which are respectively used for training and evaluating the final effect of the model. One working condition is used as source domain samples. The other working conditions are used as target domain samples. The gearbox datasets are constructed under different working conditions.
2. Model training and transfer. The convolutional neural network is constructed for training. The model weight, learning rate and other parameters are determined according to training results. The LECA module is used to extract the key features of the fault signals. Freeze the lower structure of the network, including the first convolutional layer and four Fused-MBConv modules. Fine-tune the higher structure of the network with the small samples in different working conditions. The sample distribution difference caused by different working conditions is reduced.
3. Model application. Input validation samples from the target domain into the trained model. The Softmax classification layer is used to output the results to complete the gearbox fault diagnosis for varying working conditions. Furthermore, the cross-component fault diagnosis is further realized according to the above process.

Experimental verification
Data introduction

The gearbox dataset from Southeast University is used as fault diagnosis experimental data in this paper, which was obtained on drivetrain dynamic simulator (DDS)²⁶. This platform is composed of a programmable controller, a variable speed drive motor, a two-stage planetary gearbox and a two-stage parallel shaft spur gearbox, etc. This dataset includes two sub datasets: bearing and gearbox datasets. Each sub dataset has one health state and four fault states. The detailed fault types are shown in Table 2.

The sample size of source domain is 4000, which is used to train the model. Each fault type consists of 800 samples, divided into a training set and a validation set in a 4:1 ratio. The sample size of target domain is 1000, which is used to verify the generalization and transfer effects for small samples. Each fault type consists of 200 samples, divided into a training set and a validation set in a 1:4 ratio. The detailed partitioning is shown in Table 2.

Wavelet transform is used to process vibration signals, mapping the original signal to 2D space. The wavelet time–frequency images of bearing and gear are obtained, as shown in Figs. 6 and 7.

Parameter settings

The entire experiment is performed under the ubuntu 18.04 operating system, applying Python 3.8 and Pytorch 1.8 framework. It runs on a computer of an Intel (R) Xeon(R) Gold 6330 processor and a NVIDIA GeForce RTX 3090 GPU. During the experiment, Adaptive Moment Estimation (Adam) is used to update the training parameters of all models. Cross entropy is used to calculate loss values. The dropout is set to 0.2. Batch Size affects the generalization performance and convergence speed. Too small batch size could lead to a large impact on the training process. Too large could not achieve the ideal accuracy for a limited number of epochs³⁵. Therefore, the

Subject	Type	Description	Source domain		Target domain		Working conditions
			training	validation	training	validation	
Bearing	Normal	Health state	640	160	50	150	20 Hz-0 V, 30 Hz-2 V
	Ball	Ball crack	640	160	50	150	20 Hz-0 V, 30 Hz-2 V
	Inner	Inner ring crack	640	160	50	150	20 Hz-0 V, 30 Hz-2 V
	Outer	Outer ring crack	640	160	50	150	20 Hz-0 V, 30 Hz-2 V
	Combination	Inner and outer ring crack	640	160	50	150	20 Hz-0 V, 30 Hz-2 V
Gear	Normal	Health state	640	160	50	150	20 Hz-0 V, 30 Hz-2 V
	Chipped	Teeth crack	640	160	50	150	20 Hz-0 V, 30 Hz-2 V
	Miss	Missing teeth	640	160	50	150	20 Hz-0 V, 30 Hz-2 V
	Root	Root crack	640	160	50	150	20 Hz-0 V, 30 Hz-2 V
	Surface	Surface wear	640	160	50	150	20 Hz-0 V, 30 Hz-2 V

Table 2. Bearing and gear failure types description and Dataset partitioning.

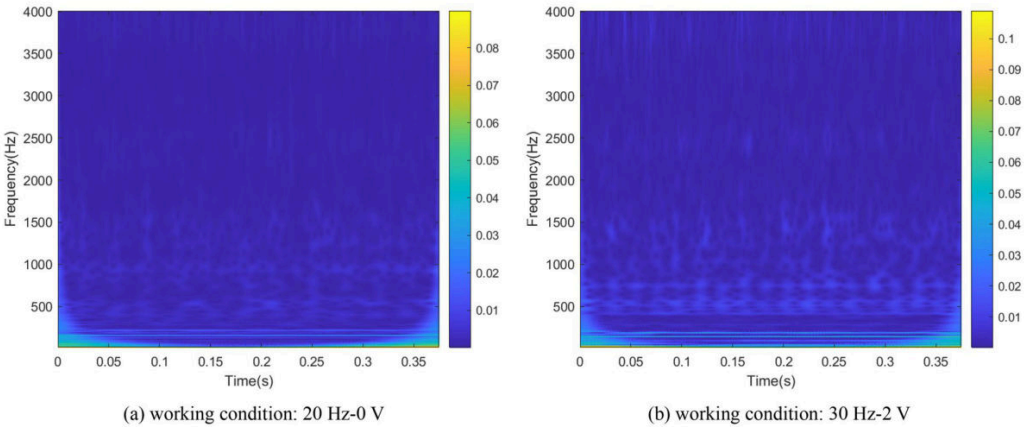


Figure 6. The wavelet time–frequency images of bearing signals.

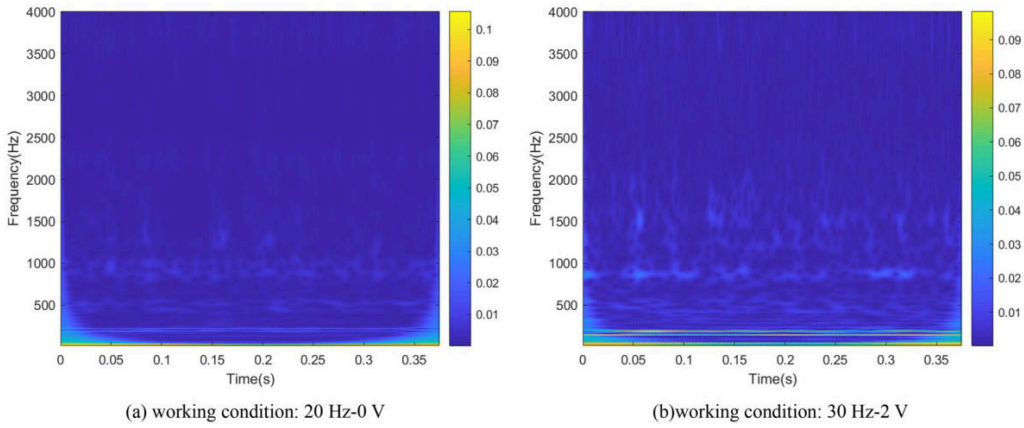


Figure 7. The wavelet time–frequency images of gear signals.

Batch Size is set to 32. In the initial stage, a higher learning rate can quickly approach the optimal solution. The learning rate decay enables the model to make large weight adjustment at the initial training stage. It can perform more precise parameter adjustments near the optimal solution in the subsequent stages. The initial learning rate is set to 0.01. The learning rate decay strategy is used to optimize training process. The accuracy of the validation

is set as an indicator. The learning rate is adjusted when the accuracy no longer rises. The experimental variable settings are shown in Table 3.

Experiment and result analysis

Experimental verification of attention mechanism

To explore the fault diagnosis effectiveness of EfficientnetV2 network integrated with LECA, the classification accuracy of EfficientNetV2 combined with SE, ECA and LECA is compared and analyzed respectively. Bearing and gear datasets at 20 Hz-0 V working conditions are used for verification. Each model is trained for 10 times to mitigate the influence of random initial values. The average is taken. The comprehensive performance of the models is shown in Table 4.

From Table 4, the accuracy of the three models is all above 90% on both datasets. The diagnostic performance of LECA-EfficientNetV2 is the best among all comparative methods. The model accuracy rises reasonably after considering the multi-scale input. A small size of convolution kernel could extract richer features. The accuracy reaches 99.38% and 99.75% respectively, an improvement of about 3% based on ECA module. This demonstrates the effectiveness of the LECA module.

In terms of fault diagnosis efficiency, SE-EfficientNetV2 has the longest diagnosis time, with an average iteration time of about 14.61 s. SE has two fully-connected layers and high computational costs. The kernel size of LECA module is smaller than ECA. The LECA-EfficientNetV2 spends the shortest diagnosis time, with about 13.57 s, as shown in Fig. 8. By comprehensive comparison, with 50 iterations, the LECA-EfficientNetV2 model has the shortest diagnosis time and the highest accuracy.

To further evaluate the superiority of the three models, Fig. 9 presents a validation accuracy curve of two experimental subjects for 50 iterations. Compared with the networks based on SE and ECA, the accuracy of LECA-EfficientNetV2 model always more than 97% after about ten iterations, which further demonstrates the model integrated with LECA module could better complete gearbox fault diagnosis.

To deeply explore the reasons for false classification of samples, the paper presents the validation results of bearing and gear data under 20 Hz-0 V working condition in the form of confusion matrix, as shown in Fig. 10.

Variables	Definitions	Values
Batch Size	Number of samples need to be processed per batch	32
Initial learning rate	Learning rate setting at the initial training	0.01
Learning rate decay	Automatically adjust learning rate by a decrease of 0.1	0.1
Dropout	Randomly delete neurons	0.2

Table 3. Experimental variable definition and settings.

	SE-EfficientNetV2 (%)	ECA-EfficientNetV2 (%)	LECA-EfficientNetV2 (%)
Bearing	94.63	95.5	99.38
Gear	95.23	97	99.75

Table 4. Bearing and gear fault diagnosis accuracy of models based on three attention mechanisms.

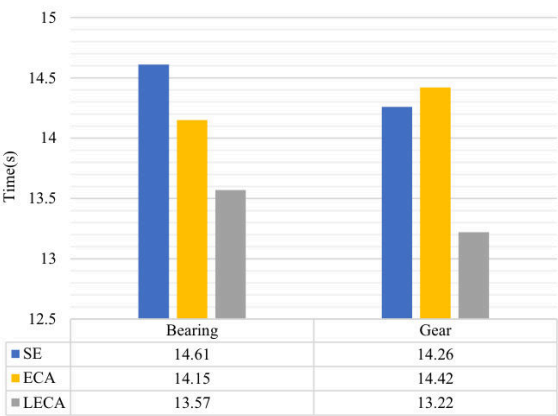


Figure 8. The fault diagnosis time for models based on three attention mechanisms.

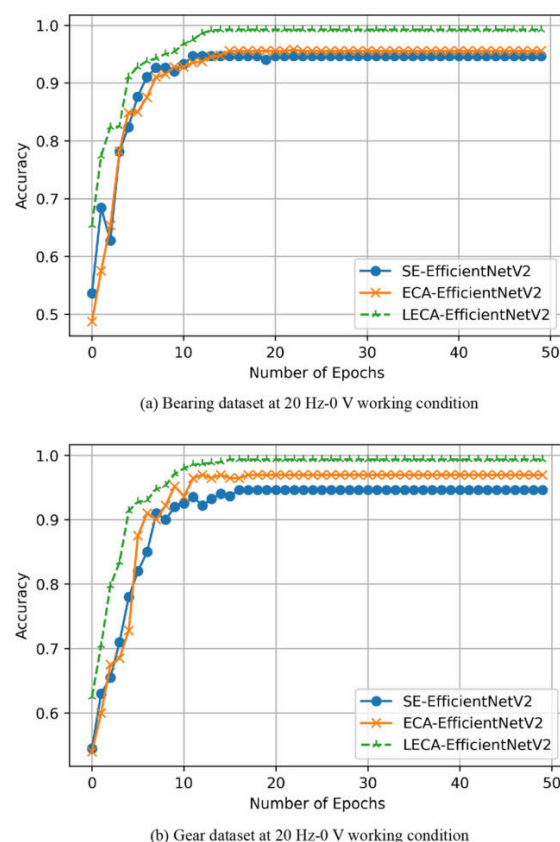


Figure 9. Accuracy curves of models based on three attention mechanisms for 50 iterations.

The recognition capacity of SE-EfficientNetV2 on Ball and Surface faults are relatively poor. The classification effect of ECA-EfficientNetV2 on Comb and Miss faults need to be improved. In contrast, LECA-EfficientNetV2 network greatly improves the recognition accuracy on different samples. Only a few samples are misclassified. Some Normal samples are misclassified as Comb on the bearing dataset. A few Miss samples are considered as Root on gear dataset. The recognition precision reaches almost 100% on other fault types. This proves once again the excellent fault feature learning capacity of LECA-EfficientNetV2 network. Therefore, the further research is to establish a transfer learning network based on the LECA-EfficientNetV2 network.

Experimental verification of transfer learning fault diagnosis

The previous experiments were conducted on a dataset with sufficient fault samples. To achieve high-accuracy fault diagnosis for untrained small samples, transfer learning method is introduced and different types of transfer learning fault diagnosis tasks are set. After the network of the network is trained on source domain dataset, the lower structures of the network are frozen and the higher structures of the network are fine-tuned with 250 target domain samples. Then 750 validation samples from target domain are used to estimate the classification capacity.

To verify the performance of the proposed network, the fault diagnosis results are compared with seven other models, namely Vgg13³⁶, ResNet50³⁷, MobileNetV3-L³⁸, EfficientnetV1-b0³⁹, EfficientnetV2-S³³, GhostNetV2⁴⁰, FasterNet-T2⁴¹ network. The above models are trained same as LECA-EfficientNetV2 network. To ensure the reliability of experimental results, the average value of 10 experiments is taken as result. The detailed transfer experiments are shown in Table 5. T1 represents the transfer of bearing fault diagnosis knowledge learned from source domain (20 Hz-0 V) to target domain (30 Hz-2 V). T3 represents the transfer of bearing diagnosis knowledge learned from source domain (20 Hz-0 V) to gear target domain (20 Hz-0 V).

The classification accuracy of eight transfer learning fault diagnosis models is shown in Fig. 11. The results present the proposed models have stronger transfer feature learning ability than other models. The accuracy is 99.27% and 99.63% in T1 and T2 respectively. It proves the model is effective in diagnosing small sample faults under variable working condition. The accuracy of T3 and T4 is 99.15% and 99.02%. This method can achieve cross-component fault diagnosis and has good generalization. Among the comparative methods, The accuracy of EfficientNet series network is worse than proposed method in four tasks. It proves the model combining multi-scale feature input and LECA module can obtain richer signal information. The accuracy of FasterNet-T2

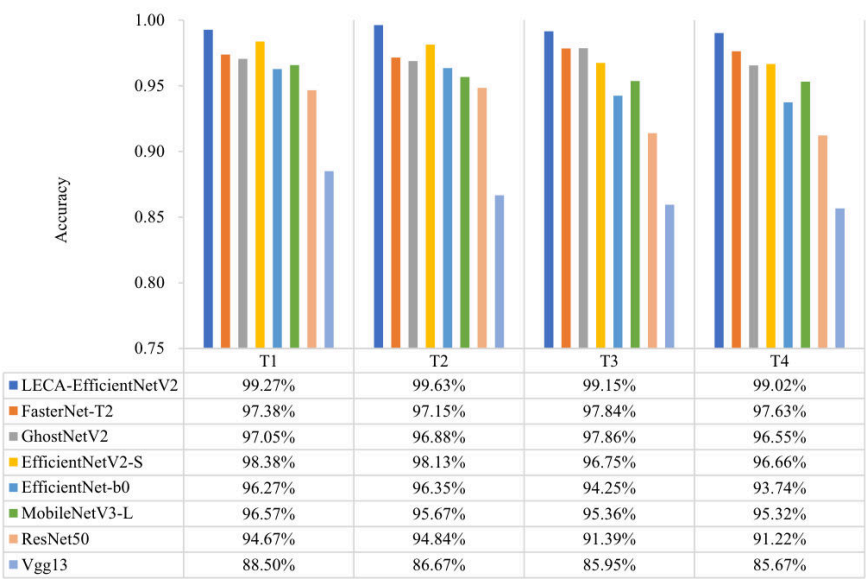


Figure 11. Classification accuracy of eight transfer learning fault diagnosis models in four tasks.

is similar to GhostNetV2, both above 97%. The Vgg13 has bad generalization ability and the worst diagnostic effect. The above results demonstrate the method can effectively extract fault features under different working conditions and components.

The fault diagnosis time of eight transfer learning models is shown in Fig. 12. The results show the proposed model achieves the shortest diagnosis time in four tasks. The diagnosis time is 9.73 s, 9.58 s, 9.92 s and 9.79 s in four tasks. It can realize the fast fault diagnosis in varying working condition and cross components. In addition to training time, Floating-point Operations (FLOPs) and Parameters (Params) are usually regarded as indicators to evaluate the complexity of the model. FLOPs presents the number of floating-point operations. Params is the number of parameters of the model³⁵. The complexity of eight network is shown in Table 6. The parameters of

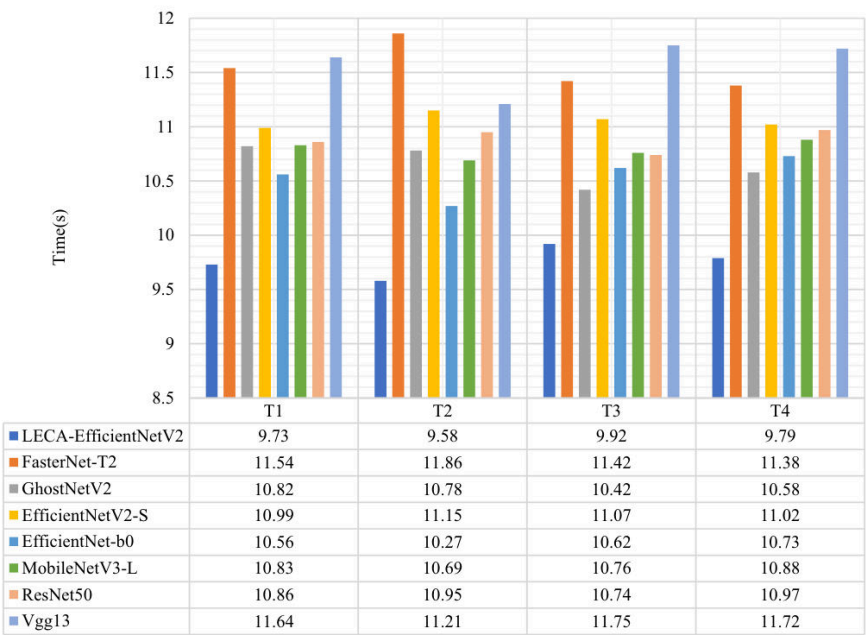


Figure 12. Fault diagnosis time of eight transfer learning fault diagnosis models in four tasks.

Model	FLOPs(G)	Params(M)
LECA-EfficientNetV2	0.98	1.19
FasterNet-T2	1.91	13.71
GhostNetV2	0.18	4.88
EfficientNetV2-S	2.88	20.18
EfficientNet-b0	0.41	4.01
MobileNetV3-L	0.37	4.21
ResNet50	4.12	23.52
Vgg13	11.38	128.98

Table 6. The FLOPs and Params of eight models.

LECA-EfficientNetV2 is the smallest, but FLOPs is not the lowest. It could be related to the depth of the network, different convolution and other parameter settings. Combined with the fault diagnosis time of the model, LECA-EfficientNetV2 meets the requirements for model lightweight.

To observe the distribution variation process of fault data intuitively, this paper uses the t-SEN method to visualize the classification process of bearing data in T1 tasks. The detailed feature distribution is shown in Fig. 13. The dimensionality reduction visualization shows, without classification, the feature distributions of various fault signals are obviously mixed and difficult to distinguish. With the further training of the model, there are already relatively obvious five kinds of distributions in the fully-connected layer.

Conclusion

This paper proposes a new gearbox fault diagnosis method based on lightweight channel attention mechanism and transfer learning. The method could solve the problem of bad fault diagnosis performance caused by large sample distribution difference and limited samples. The bearing and gear datasets are used to verify the classification and generalization capacity of the proposed model. The conclusions are as follows.

(1)LECA-EfficientNetV2 has been proven to get 99.38% and 99.75% accuracy on bearing and gear samples, respectively. The fault diagnosis time is 13.57 s and 13.22 s. Compared with SE-EfficientNetV2 and ECA-EfficientNetV2, LECA-EfficientNetV2 has the best diagnostic effect on both datasets. It could extract more detailed features and effectively complete gearbox fault diagnosis.

(2)The transfer learning experiments present LECA-EfficientNetV2 has the best diagnostic performance and generalization ability under different gearbox working conditions and components. The computational cost shows that proposed method could meet the requirement for model lightweight. The proposed method can realize fast and accurate classification of gearbox faults. It is of great significance to solve the problem of small samples in practical engineering applications.

Since this paper only explores two components and the validation datasets are completely balanced. However, in industrial environment, the sample imbalance problem is prominent. The aspects will be further explored in the future: (1)Further expand the application scope of LECA-EfficientNetV2 to enhance model generalization ability. (2)Further study the model performance in imbalanced datasets and maintain high accuracy.

extractor. This network consists of four hierarchical stages (Stage 1–4), each comprising patch partitioning (with a patch size of 4×4), linear embedding (embedding dimension = 96), window-based multi-head self-attention (window size = 7), and multi-scale feature aggregation. The final output is a global image-level feature vector with a dimensionality of 768, which serves as input for subsequent multimodal fusion.

VMD Channel: To enhance the model's capability in representing local mode components within non-stationary signals, the sequence branch applies Variational Mode Decomposition (VMD) followed by a lightweight hybrid modeling module based on CNN and BiGRU. VMD decomposes each input signal into $K=4$ Intrinsic Mode Functions (IMFs), which are stacked into a multi-channel input tensor of shape (4, 1024). This structure preserves distinct frequency components and facilitates the modeling of complementary modal information.

The multi-channel tensor is processed through a feature extraction backbone consisting of three one-dimensional convolutional layers followed by a single-layer bidirectional gated recurrent unit (BiGRU). The convolutional block performs convolution, non-linear activation (ReLU), and max pooling operations to capture localized temporal patterns. The resulting sequence is fed into the BiGRU module, which models bidirectional temporal dependencies. An attention mechanism is then applied to perform weighted aggregation over the BiGRU outputs, producing a sequence-level feature vector with a dimensionality of 128. This architecture is designed to effectively extract and integrate temporal features for industrial fault diagnosis tasks.

To further evaluate the computational cost of each module in the dual-channel fault diagnosis framework, we computed the parameter counts and floating-point operations (FLOPs) for Swin Transformer, CNN, BiGRU, Attention, and the MCB fusion module. The results are summarized in Table 1.

It is evident that the Swin Transformer constitutes the majority of both parameters and FLOPs, while the VMD branch modules contribute relatively minor computational load. This distribution is consistent with the design rationale: the CWT branch, processed by the Swin Transformer, extracts rich hierarchical spatial features from time-frequency images, which are critical for accurate fault discrimination. Meanwhile, the VMD branch provides complementary temporal features through CNN, BiGRU, and attention, requiring relatively less computation. Therefore, the Swin Transformer dominating the overall computational complexity is reasonable and aligned with its primary role in capturing the most informative fault-related patterns.

The overall fault diagnosis procedure of the SCBM-Net model is illustrated in Algorithm 1 and Fig. 2. The detailed steps are as follows:

Step 1: The original vibration signals are preprocessed using a sliding window technique. Overlapping time series segments of equal length are generated with a fixed stride, ensuring temporal continuity between samples. The dataset is then divided into training and testing sets according to a predefined ratio to maintain consistency and balance.

Step 2: Continuous Wavelet Transform (CWT) is applied to the preprocessed time series signals, converting one-dimensional vibration data into two-dimensional time-frequency representations. CWT effectively preserves the time-frequency dependency of the signals, maintains the integrity of the features, and introduces spatial characteristics that enhance classification performance. Additionally, the relatively small size of the CWT-generated images reduces storage and computational costs.

Step 3: Variational Mode Decomposition (VMD) is employed to decompose the raw signals into a set of Intrinsic Mode Functions (IMFs). This adaptive decomposition enhances the discriminability of fault-related features and effectively suppresses noise interference. VMD allows for the extraction of key features across multiple temporal scales, improving the separability of different modal components in the feature space.

Step 4: A multimodal dataset is constructed, ensuring the consistency of inputs between the two channels. The time-frequency images generated by CWT and the sequential tensor data obtained from VMD are aligned via a strict sample ID matching mechanism. This guarantees synchronization of input data across both channels and enables joint feature extraction from time-frequency images and time-domain signals.

Step 5: A parallel dual-channel deep learning model is constructed. In Channel 1, a Swin Transformer is used to extract features from the CWT time-frequency images. Its hierarchical representation structure and shifted window self-attention mechanism allow effective modeling of both local and global information, enhancing the network's capability to express image features. In Channel 2, a CNN-BiGRU-Attention network is employed to extract features from the VMD-decomposed sequential signals. CNN captures local temporal patterns, BiGRU models temporal dependencies in both forward and backward directions, and the attention mechanism adaptively emphasizes key fault-related features while suppressing irrelevant information.

Step 6: The features extracted from both channels are fused to enhance fault classification performance. A Multimodal Compact Bilinear (MCB) pooling module is used to deeply integrate features from both channels, achieving complementary enhancement of multimodal information. Adaptive average pooling and dimensionality reduction are applied to reduce computational complexity. In the final classification stage, a fully connected layer followed by a Softmax classifier is used to identify fault types. The effectiveness and robustness of the model are validated through experimental evaluation metrics and visualization analyses.

Module	Parameters	FLOPs
Swin transformer	86.87 M	15466.9 M
CNN	31.3 K	19.66 M
BiGRU	494.6 K	1346.37 M
Attention	98.7 K	134.48 M
MCB	0	57.34 K

Table 1. Model complexity analysis of each module (parameters and FLOPs).

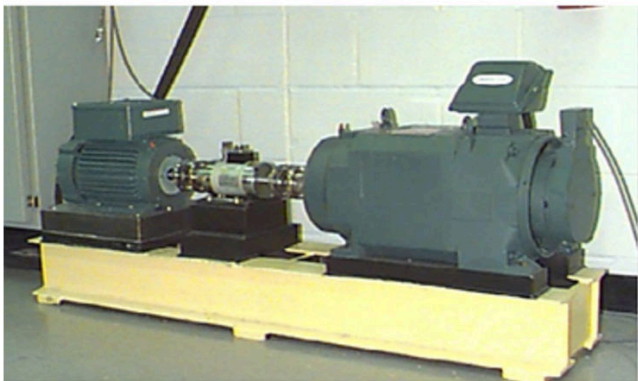


Fig. 3. CWRU bearing vibration dataset experimental platform.

Fault type	Label	Damage diameter (mil)	Training set	Test set
N	0	–	140	60
IR007	1	7	140	60
B007	2	7	140	60
OR007	3	7	140	60
IR014	4	14	140	60
B014	5	14	140	60
OR014	6	14	140	60
IR021	7	21	140	60
B021	8	21	140	60
OR021	9	21	140	60

Table 2. Public bearing dataset collected by CWRU.

To eliminate the possibility of cherry-picking, all critical hyperparameters were optimized through systematic grid search combined with validation performance monitoring. Candidate values for the learning rate were set to {1e-5, 1e-4, 1e-3}, while batch sizes of {16, 32, 64} were tested. The optimal configuration was determined based on the highest average F1-score on the validation set.

Model training was performed with the Adam optimizer and weight decay for regularization. A StepLR scheduler was applied to reduce the learning rate by a factor of 0.1 every 10 epochs, thereby facilitating stable convergence. The batch size was set to 32, which provided a good balance between convergence stability and GPU memory efficiency. Early stopping with a patience of 5 epochs on the validation accuracy was used to prevent overfitting, and dropout ($p=0.2$) was applied in the BiGRU layers. The cross-entropy loss was employed as the optimization objective. Training proceeded for up to 50 epochs, and the model parameters that achieved the best validation accuracy were preserved for testing.

To ensure reproducibility, all experiments were conducted with fixed random seeds applied to PyTorch, NumPy, and Python's random module, and deterministic behavior was enforced by disabling non-deterministic CUDA operations. The main implementation environment included Python 3.10, PyTorch 2.6.0 with CUDA 12.4, and torchvision 0.21.0 with CUDA 12.4. All critical training and model parameters, including optimizer settings, learning rate schedule, batch size, number of epochs, early stopping criteria, regularization techniques, model dimensions, and random seed, are summarized in Table 3, providing sufficient information for replication of all reported results.

VMD data processing

The continuous one-dimensional time-series vibration signals are segmented into multiple samples of fixed length, each containing 1024 data points, using a sliding window approach with a 50% overlap rate. Each sample is assigned a unique global sample ID and its corresponding fault label. To ensure data balance across different fault categories, 200 samples are randomly selected from each fault type. These are further divided into 140 training samples and 60 testing samples per category, ensuring that the dataset satisfies the experimental requirements.

In this study, the Variational Mode Decomposition (VMD) algorithm is employed to preprocess the original signals, which are subsequently converted into PyTorch tensors. For each sample, the VMD algorithm adaptively decomposes the signal into a series of Intrinsic Mode Functions (IMFs). Taking the IR007 fault signal as an example, the time-domain decomposition results are illustrated in Fig. 4(a). The original signal is decomposed

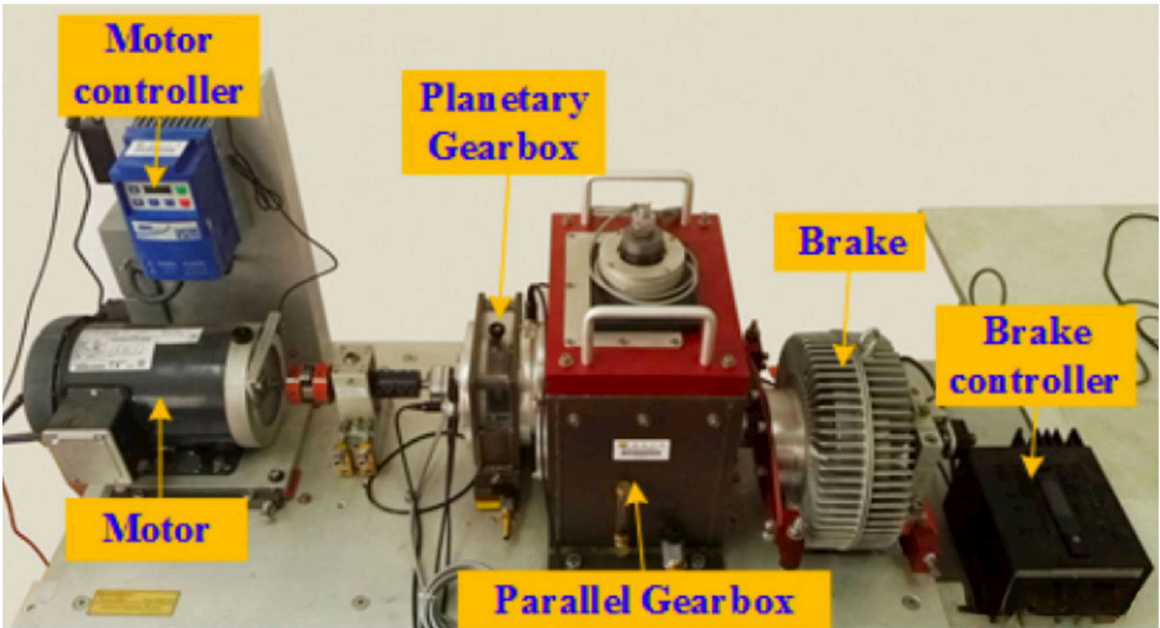


Fig. 13. Bearing test rig structure at Southeast University.

Fault type	Label	Speed/load	Training set	Test set
Ball_0	0	20/0	210	90
Ball_1	1	30/2	210	90
Comb_0	2	20/0	210	90
Comb_1	3	30/2	210	90
Health_0	4	20/0	210	90
Health_1	5	30/2	210	90
Inner_0	6	20/0	210	90
Inner_1	7	30/2	210	90
Outer_0	8	20/0	210	90
Outer_1	9	30/2	210	90

Table 10. Publicly available bearing dataset from Southeast University.

Few-shot experiment

To further evaluate the recognition capability of SCBM-Net under limited data conditions, a few-shot experiment was conducted. Specifically, the experiment was based on the CWRU bearing fault dataset. For each fault category, only 60 samples were randomly selected for training, while the remaining 140 samples were used for testing. The detailed sample distribution is shown in Table 12. This setting simulates practical industrial scenarios where fault data is scarce. All other experimental configurations were kept consistent with the full-sample setting to ensure fair comparability.

To better illustrate the classification effectiveness of the SCBM-Net model, the t-Distributed Stochastic Neighbor Embedding (t-SNE) method is employed to visualize the feature distributions on the test set. Specifically, the original data distribution, the classification results of the two single-channel branches, and the final classification results of SCBM-Net are presented separately, as shown in Fig. 15.

As shown in Fig. 15(a), the original data distribution is chaotic and disordered, lacking any clear clustering structure. After feature extraction through the VMD channel (Fig. 15(b)) and the CWT channel (Fig. 15(c)), the samples exhibit a certain degree of clustering tendency and class separation, but there is still considerable overlap between different classes, resulting in limited classification performance. In contrast, Fig. 15(d) shows that the SCBM-Net model, after fusing time-frequency features, significantly improves the class discrimination ability. The samples are distributed more compactly and with clearer boundaries in the feature space, successfully achieving effective differentiation of 10 types of bearing faults. Under small sample conditions, the model can still reach a classification accuracy of 98.64%, fully demonstrating its superior feature representation capability and fault diagnosis performance.

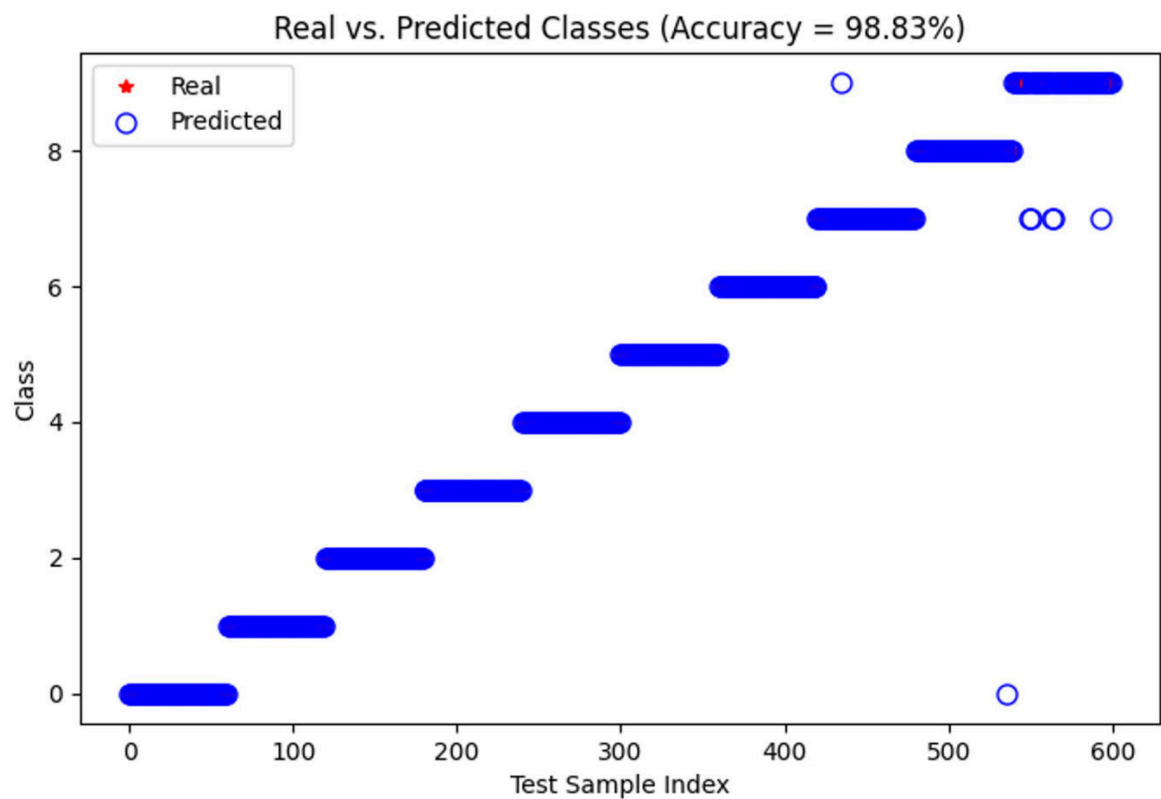


Fig. 14. Experimental results based on the Southeast University bearing dataset.

Model	Dataset		Accuracy (%)	Average accuracy (%)
	CWRU	SEU		
VMD + CNN	√		82.00%	56.78%
		√	31.56%	
VMD + CNN-BiGRU	√		87.50%	66.31%
		√	45.11%	
VMD + CNN-BiGRU-Attention	√		97.50%	76.42%
		√	55.33%	
CWT + Swin transformer	√		99.33%	98.50%
		√	97.67%	
SCBM-Net	√		99.83%	99.08%
		√	98.33%	

Table 11. Experimental accuracy comparison.

Unbalanced dataset experiment

To further verify the robustness and generalization capability of the proposed SCBM-Net model under practical complex working conditions, a comparative experiment based on an unbalanced dataset was designed and conducted. In industrial equipment operation, the distribution of fault categories typically exhibits significant imbalance, where some fault types occur less frequently and thus constitute minority classes. It is necessary to evaluate the model's ability to recognize various fault types under such imbalanced data distributions.

The experimental data is constructed based on the CWRU bearing fault dataset. To simulate an imbalanced scenario, four major fault categories (Normal, IR007, IR014, IR021) each retain 200 samples, with 140 samples for training and 60 for testing; the remaining six minor categories (B007, OR007, B014, OR014, B021, OR021) are each limited to 45 samples, divided into 30 for training and 15 for testing. The detailed composition of this unbalanced dataset is shown in Table 13.

Method	None	10 dB	5 dB	0 dB	-5 dB	-10 dB
Textual Method	99.83%	99.50%	99.17%	97.00%	93.67%	80.67%
ACNN-LFSwinT	97.10%	93.30%	91.50%	88.90%	85.20%	79.70%
CNN-BiLSTM	96.46%	92.84%	90.23%	77.34%	73.33%	70.77%
MLF-MSI	99.33%	99.00%	95.50%	93.30%	87.23%	75.17%
MS-Residual	98.67%	97.50%	96.67%	94.83%	82.17%	69.50%
MSDCSA	94.50%	88.50%	85.00%	79.17%	76.67%	70.83%
TST	90.00%	88.17%	85.33%	82.67%	75.83%	75.00%
Feat-ML	95.67%	93.67%	92.83%	89.00%	81.33%	75.33%

Table 14. Accuracy of fault diagnosis methods under different SNR levels (%).

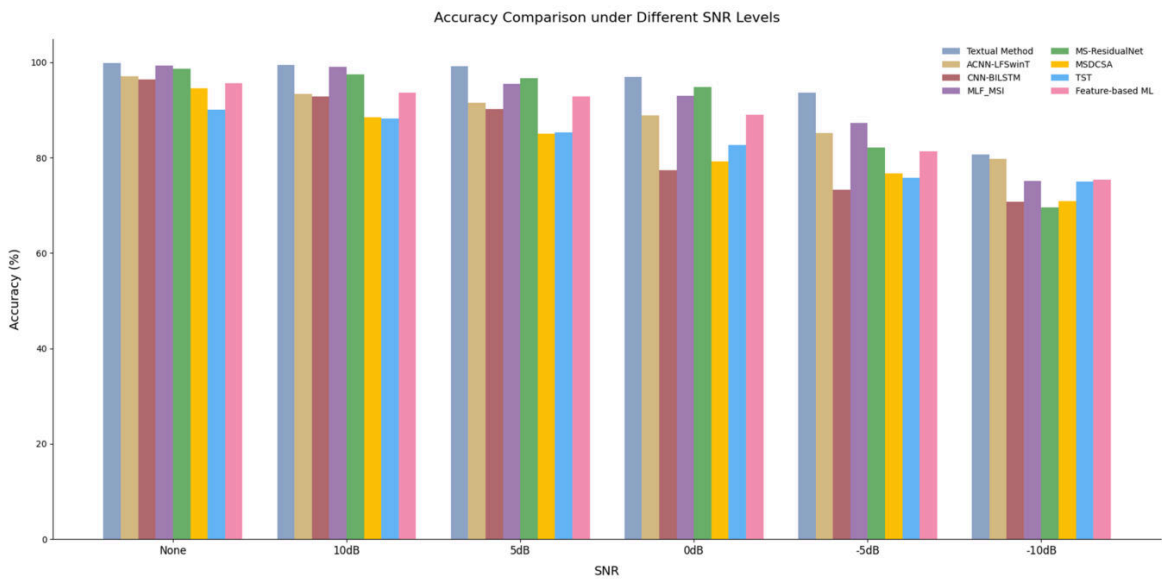


Fig. 19. Comparison of the proposed model with other fault diagnosis models.

(CWT) and Variational Mode Decomposition (VMD) into a unified architecture. Specifically, a Swin Transformer is employed in the image channel to extract both local and global features from CWT-based time–frequency images, while a CNN-BiGRU-Attention network captures temporal dependencies from the intrinsic mode functions generated by VMD in the sequential channel. These heterogeneous features are effectively fused via a Multimodal Compact Bilinear (MCB) pooling module, substantially enhancing the model’s feature representation capability.

Extensive experiments conducted on the CWRU and Southeast University datasets validate the proposed model’s superior diagnostic accuracy, generalization performance, and robustness to noise. SCBM-Net achieves 99.83% accuracy under noiseless conditions, 98.33% in cross-domain evaluations, and maintains 80.67% accuracy at a low signal-to-noise ratio of –10 dB. Additionally, it demonstrates strong performance on imbalanced datasets, accurately identifying both majority and minority fault classes. Ablation studies further confirm the contribution of each core module to the overall performance gains. Furthermore, under small-sample conditions, SCBM-Net still achieves a high classification accuracy of 98.64%, highlighting its excellent feature representation capability and adaptability in data-scarce scenarios.

Data availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Operation	Input shape	Kernel size	Stride size	Output shape
Input layer	64 × 64 × 3	–	–	–
Convolutional layer	64 × 64 × 3	5 × 5 × 32	1	64 × 64 × 32
Pooling layer	64 × 64 × 32	2 × 2	2	32 × 32 × 32
Convolutional layer	32 × 32 × 32	3 × 3 × 64	1	32 × 32 × 64
Pooling layer	32 × 32 × 64	2 × 2	2	16 × 16 × 64
IResNet_A	16 × 16 × 64	(3 × 3) / MHA(4)	1/0	16 × 16 × 64
Pooling layer	16 × 16 × 64	2 × 2	2	8 × 8 × 64
IResNet_B	8 × 8 × 64	(1 × 1) / MHA(4)/(1 × 1)	1/0/1	8 × 8 × 128
Pooling layer	8 × 8 × 128	2 × 2	2	4 × 4 × 128
IResNet_B	4 × 4 × 128	(1 × 1) / MHA(4) / (1 × 1)	1/0/1	4 × 4 × 256
Pooling layer	4 × 4 × 256	2 × 2	2	2 × 2 × 256
Fully-connected layer	1024	–	–	256
Fully-connected layer	256	–	–	10

Table 1. Model parameter settings of IResNet.

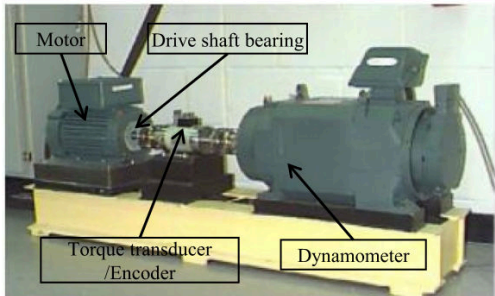


Fig. 9. CWRU rolling bearing experimental platform by⁴⁰.

a classifier to facilitate fault diagnosis. The fault diagnosis model is trained using a batch size of 32 and over 30 epochs. Cross-entropy loss is utilized as the error loss function, with Adam as the optimizer. The learning rate is set at 0.001. The learning rate attenuation mechanism is employed to fine-tune the learning rate, optimize training effectiveness, and minimize training costs. The model parameter settings are shown in Table 1.

Experimental verification

To verify the effectiveness and superiority of the method proposed in this paper, experimental verification has been carried out for two bearing datasets: the bearing dataset of Case Western Reserve University⁴⁰, the bearing dataset of Southeast University⁴¹ and the bearing dataset of Jiangnan University⁴².

Case 1: CWRU bearing dataset

Experiment setup and data preprocess

The experimental dataset was sourced from the Case Western Reserve University (CWRU) Bearing Data Center. The experimental equipment is shown in Fig. 9, and the test bench is composed of a motor, rolling bearings, a torque transducer, and a dynamometer⁴⁰. The test bearing model are SKF6205 motor bearings. Faults in the bearing, including the inner ring (IR), outer ring (OR), and ball (BD), have been machined through Electrical Discharge Machining (EDM). The fault diameters are 0.1778 mm, 0.3556 mm, and 0.5334 mm for each kind of fault bearing, respectively. During the experiments, the motor was subjected to various load conditions: 0HP, 1HP, 2HP and 3HP, the sampling frequency was set at 12 kHz.

The signal is intercepted by overlapping sampling. The overlapping sampling operation formula is as follows:

$$N = \frac{L - L_1}{d} + 1 \tag{18}$$

where N is the total number of one-class samples, L is the total length of the one-dimensional signal, L_1 is the length of each signal segment, and d is the step length determining the degree of overlap between consecutive segments. The statistics of the dataset are described in Table 2. The dataset contains 10 fault categories, and the number of data samples for each fault category is 1000, including 800 training samples, 100 validation samples, and 100 test samples. Additionally, the number of data points for each data sample is 1024.

Types	Condion	Fault size(mm)	Length	Train/Valid/Test
0	IR007	0.18	1024	800/100/100
1	IR014	0.36	1024	800/100/100
2	IR021	0.53	1024	800/100/100
3	OR007	0.18	1024	800/100/100
4	OR014	0.36	1024	800/100/100
5	OR021	0.53	1024	800/100/100
6	B007	0.18	1024	800/100/100
7	B014	0.36	1024	800/100/100
8	B021	0.53	1024	800/100/100
9	Normal	0	1024	800/100/100

Table 2. The experiment dataset.

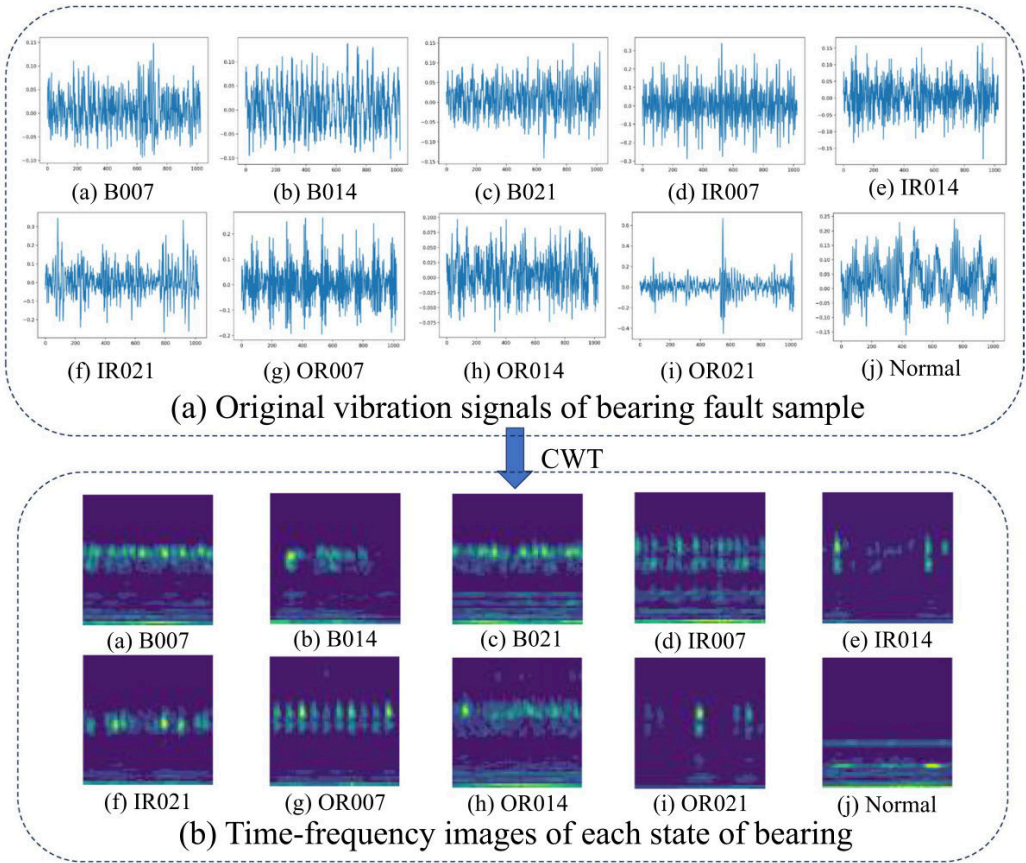


Fig. 10. 1D vibration signals are converted into 2D time–frequency images.

The original vibration signal is converted into a two-dimensional time–frequency diagram by CWT, and the results are shown in Fig. 10. After CWT, the time–frequency domain characteristics of the signal are fully obtained, which is conducive to subsequent data expansion and fault diagnosis processes.

Experimental environment and training parameter configuration

The experiments were conducted on a Windows 10 operating system with a Tesla T4 GPU, utilizing the PyTorch deep learning framework. During training, the input image resolution was set to 64×64 , with a batch size of 32 and 30 training epochs. The initial learning rate was 0.001, and the cross-entropy loss function was employed as the optimization objective. The Adam optimizer was adopted, coupled with StepLR scheduler to dynamically adjust the learning rate: the learning rate was halved every 10 epochs.

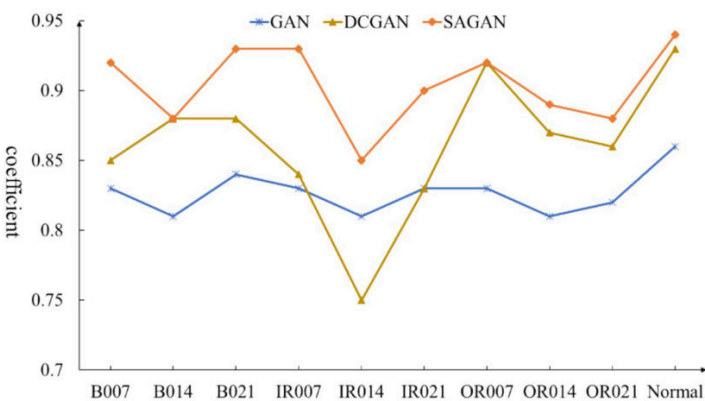


Fig. 13. Comparison of image correlation coefficients.

No	Real images	Generated image	Validation set	Testing set	Accuracy/%
1	8000	0	1000	1000	99.900
2	0	500	1000	1000	92.567
3	0	1000	1000	1000	95.233
4	0	2000	1000	1000	95.733
5	0	4000	1000	1000	97.100
6	0	6000	1000	1000	98.467
7	0	8000	1000	1000	98.500
8	8000	500	1000	1000	99.920
9	8000	1000	1000	1000	99.925
10	8000	2000	1000	1000	99.933
11	8000	4000	1000	1000	99.966
12	8000	6000	1000	1000	100
13	8000	8000	1000	1000	100

Table 3. Accuracy of diagnosis comparison under different datasets.

1 indicates a strong linear relationship. The Pearson correlation coefficient is used to analyze the correlation between the generated sample and the original sample. The analysis results are shown in Fig. 13. The analysis illustrates that SAGAN's sample generation capabilities are substantially superior to those of both standard GAN and DCGAN, highlighting its effectiveness in producing high-quality, realistic samples.

In order to verify the influence of the number of generated images on the accuracy of bearing fault diagnosis, the experiment is divided into three categories: the first category does not perform data expansion, and only uses the original data for model training (experiment 1); the second category only uses the generated data for training (experiment 2–7); the third category uses data expansion to combine the original data with the generated data to jointly train the model (experiment 8–13). The experimental results are shown in Table 3. It can be seen from the table that when the original data and the generated data are combined, the diagnostic accuracy is significantly improved. Notably, there is a clear trend showing that accuracy increases as more generated images are added to the training dataset. Specifically, when 6000 generated images are added to 8000 original samples, the accuracy rate reaches 100%. Based on these findings, subsequent experiments were conducted based on the original sample size of 8000 and the generated sample size of 6000 to further validate the benefits of this approach.

The generated data and the original data are used as the input of the CNN fault diagnosis model to train the network. The training process is shown in Fig. 14. It can be seen from the figure that after data expansion, the convergence speed of model training is obviously accelerated and the convergence is more stable. Therefore, using data generation technology to train the model by combining real data with generated data can significantly improve the stability of model training and the accuracy of bearing fault diagnosis.

The influence of network depth on diagnostic accuracy

In actual industrial production, bearing environments are typically intricate, and signals collected are often compromised by different types of noise. Therefore, varying degrees of Gaussian white noise are added to the original signal to simulate noise pollution in the industrial environment. The signal-to-noise ratio (SNR) serves as a benchmark for assessing noise intensity, which is defined as follows:

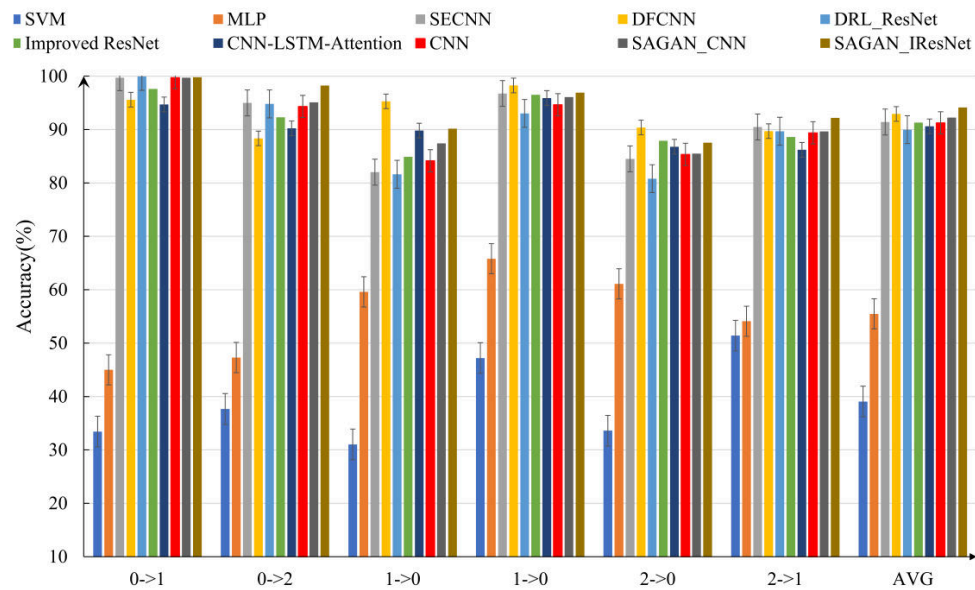


Fig. 22. The average accuracy of different models in the variable load domain.

Types	Load	Condion	Length	Train/Valid/Test
0	20 Hz-0 V	20_ball	1024	800/100/100
1	20 Hz-0 V	20_comb	1024	800/100/100
2	20 Hz-0 V	20_inner	1024	800/100/100
3	20 Hz-0 V	20_normal	1024	800/100/100
4	20 Hz-0 V	20_outer	1024	800/100/100
5	30 Hz-2 V	30_ball	1024	800/100/100
6	30 Hz-2 V	30_comb	1024	800/100/100
7	30 Hz-2 V	30_inner	1024	800/100/100
8	30 Hz-2 V	30_normal	1024	800/100/100
9	30 Hz-2 V	30_outer	1024	800/100/100

Table 4. The experiment dataset of Southeast University.

Case 2: Southeast university gearbox dataset

Experiment setup and data preprocessing

The gear data set of Southeast University comes from the open source data of Southeast University. The test bench is composed of a variable speed motor, motor controller, planetary gearbox, parallel shaft gearbox, brake, and brake controller⁴¹. The vibration signal is measured by a driveline dynamic simulator (DDS). The fault types are divided into five types: normal, ball fault, comb fault, inner ring fault and outer ring fault. These fault types are artificially set in the laboratory and can simulate the basic faults in the machinery. Regarding the operational parameters of the equipment, the speed-load settings vary between 20 Hz at 0 V and 30 Hz at 2 V.

The data is processed using overlapping sampling, with each fault category represented by 1,000 single-class samples, each with a signal segment length of 1,024. The dataset is organized into training, validation, and test sets with a distribution ratio of 8:1:1. This allocation is detailed in Table 4, which provides a statistical description of the dataset. The vibration signal is converted into a time–frequency image using CWT, and the results are shown in Fig. 23.

Experimental results and performance comparison

The 8000 original training samples and 6000 pseudo-samples are used to train the classification model. To mitigate the effects of anomalies and ensure the robustness of the results, each experiment was conducted five times. The average accuracy from these trials was computed and compared with that achieved by the improved method. The findings are presented in Fig. 24, which clearly shows that the SAGAN_IResNet method yields the highest accuracy among the tested approaches. By improving the fault classification model, the residual network is combined with the Multi-head Self-Attention mechanism to learn the sample features to the greatest extent, and the model performance is optimized as much as possible.

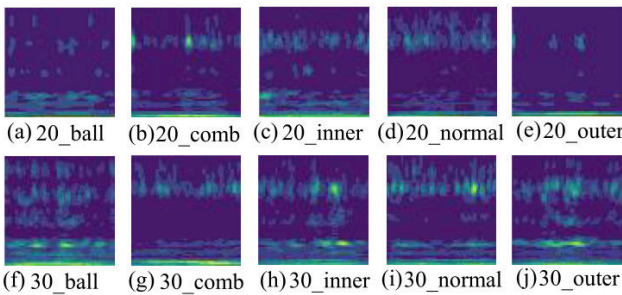


Fig. 23. Time–frequency images of ten fault categories.

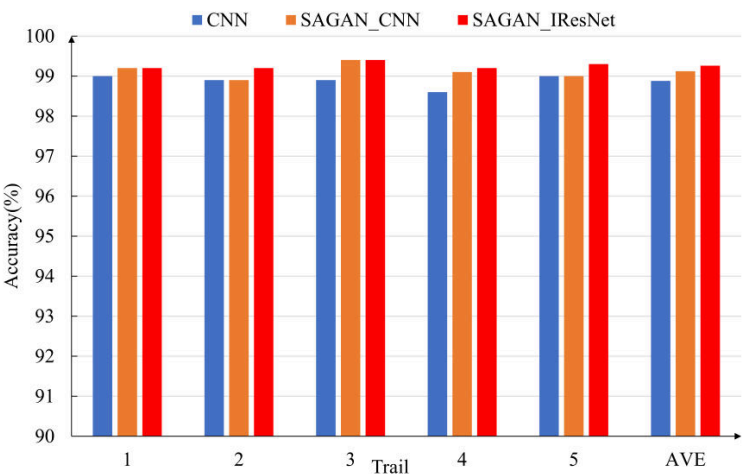


Fig. 24. Classification accuracy of the five trials in testing samples.

The detailed performance comparison of different diagnostic methods in a noisy environment is shown in Fig. 25. As can be seen from the figure, the method proposed in this paper demonstrates excellent robustness and anti-interference performance. When the signal-to-noise ratio (SNR) is 2, the fault diagnosis accuracy of this method reaches as high as 97.2%. Compared with other methods, there is a significant improvement in accuracy, which fully demonstrates its outstanding advantages and application value in a noisy environment.

Displayed in Fig. 26 is the comprehensive confusion matrix for the SAGAN_IResNet model under SNR conditions of 0 and 6. Notably, at an SNR of 0, this method yields an impressive average fault diagnosis rate of 95%. Similarly, under an SNR of 6, the model’s fault accuracy reaches 98.42%, showcasing its ability to adeptly learn fault features and accurately identify health statuses.

The fault feature learning results of the proposed method are shown in Fig. 27, which showcases the effects of increasing network depth on the classification of fault features. Initially, after CNN layer feature extraction, the results are visualized in Fig. 27a, the samples have a tendency to aggregate, but the clustering effect remains suboptimal, indicating some overlap and unclear boundaries between different fault categories. By superimposing the IResNet module, the number of network layers is gradually deepened, as shown in Fig. 27b, which significantly expands the distribution distance of fault features among different health conditions and makes the feature classification boundary clear and distinguishable.

Case 3: Jiangnan university bearing dataset

Experiment setup and data preprocessing

To verify the universality of the method proposed in this paper, the bearing dataset of Jiangnan University is used for verification⁴². This data was collected under the conditions of a sampling frequency of 50 kHz and rotation speeds of 600 r/min, 800 r/min, and 1000 r/min. The bearing dataset with a frequency of 50 kHz and a rotation speed of 800 r/min is adopted as the experimental data. According to the fault location, the bearing data is divided into four states: normal (N), inner ring (IR), outer ring (OR), and rolling element (TB). The signals are intercepted by means of overlapping sampling. The number of samples of a single class is 1000, and the signal length is 1024. The ratio of the training set, verification set, and test set is 8:1:1. The parameters of

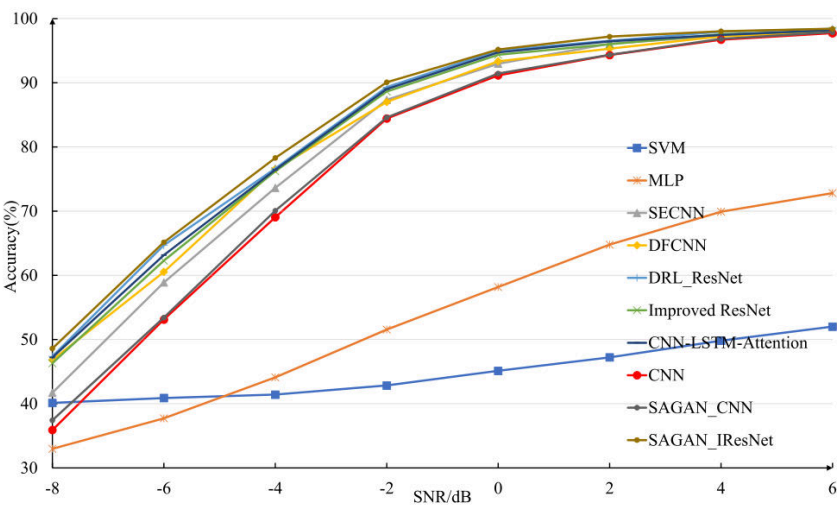


Fig. 25. The average accuracy of different models in noisy environment.

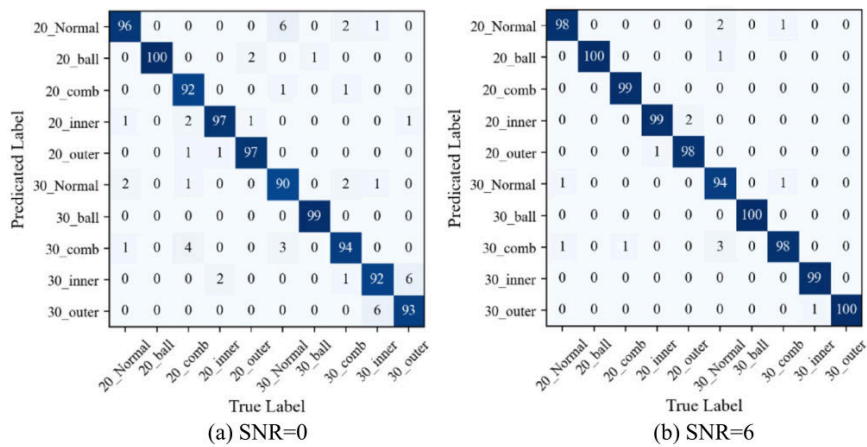


Fig. 26. Confusion matrix of diagnostic results in different noise environments.

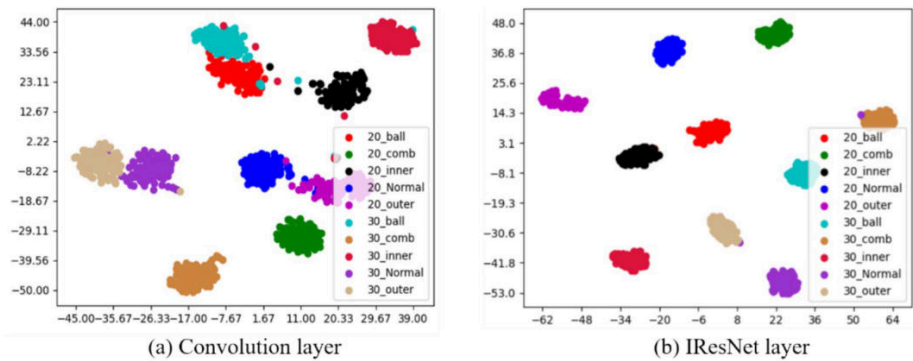


Fig. 27. Visualization of feature vectors at different positions.